CovertTux: A Cache Policy-oblivious Timing Covert Channel on a Linux System

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Introduction

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- A sender and a receiver agree on a predefined protocol and exploit the latency difference between a LLC hit and a LLC miss to communicate covertly.
- The LLC is typically shared across various cores of a chip-multiprocessor. So they are scheduled on two different cores but use the shared LLC to establish the covert channel.

Overview

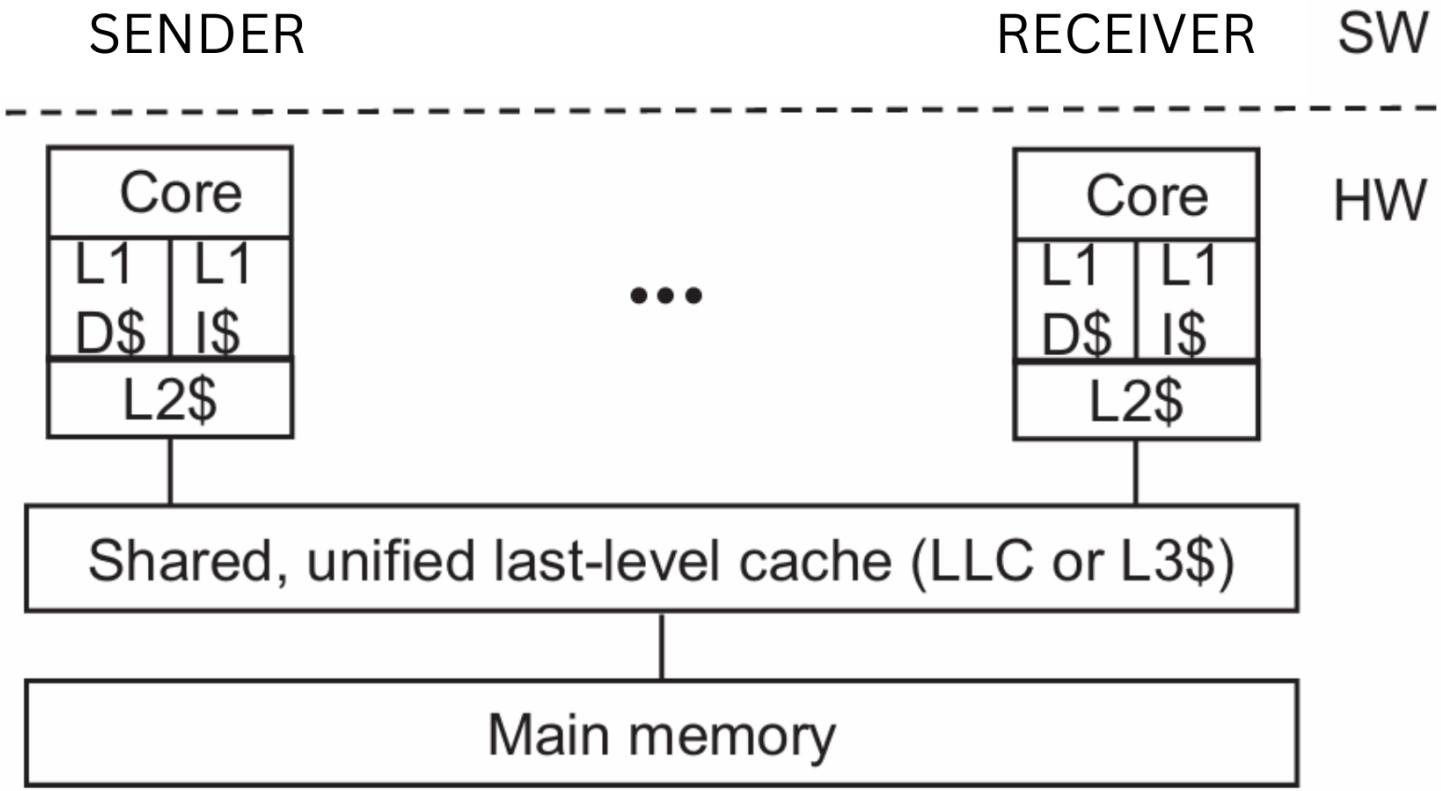

Figure: Covert channel in a real system[1]

[1] Source : https://ieeexplore.ieee.org/document/7163050

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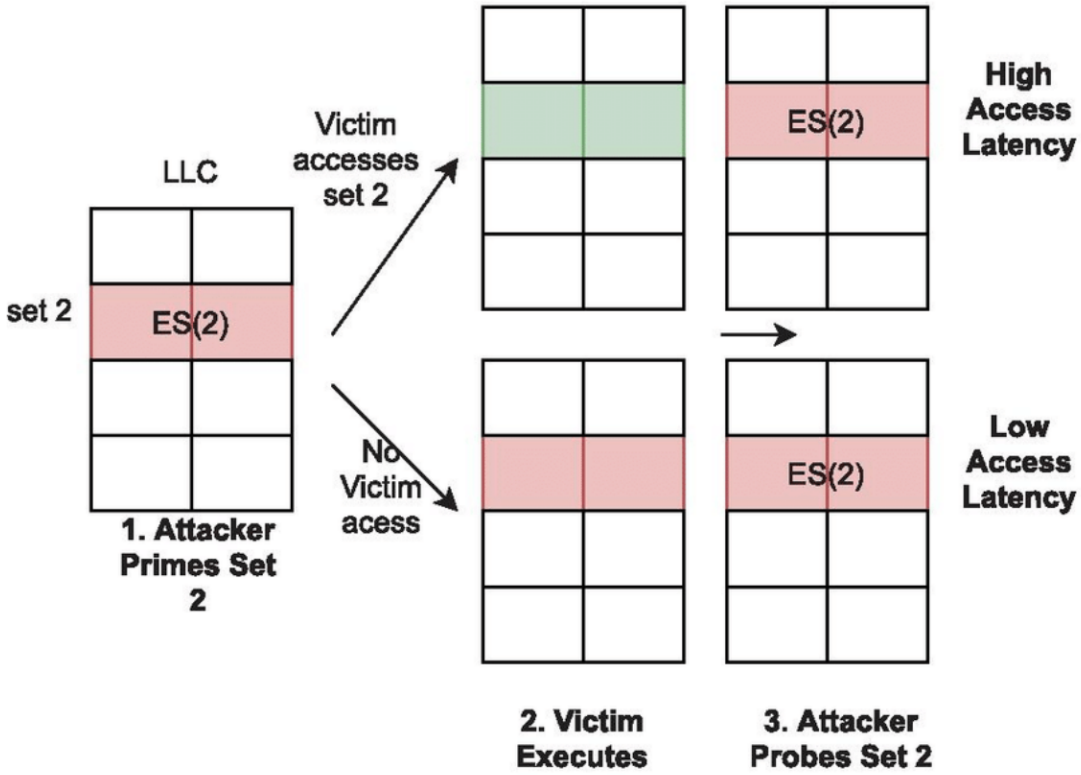

Figure: Prime+Probe Attack [2]

[2] Source : https://www.researchgate.net/publication/351710688_A_Survey_on_Cache_Timing_Channel_Attacks_for_Multicore_Processors

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- Any replacement policy would work.
- A set of a set-associative LLC works.

Our Machine Configurations

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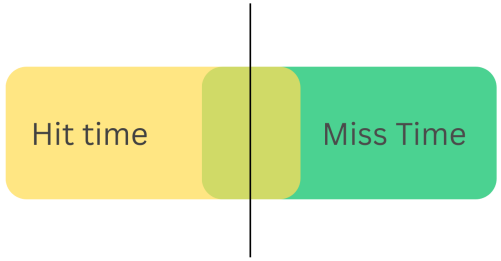
- We used one of our CSE server machines - 172.27.21.1
- It is a Intel(R) Xeon(R) E-2278G CPU running at 3.40GHz having 16 logical cores
- It has a three level cache hierarchy.
- L1i Cache : 32 KB, 8-ways
- L1d Cache : 32 KB, 8-ways
- L2 Cache : 256 KB, 4-ways, non-inclusive
- L3 Cache (LLC) : 16 MB, 16-ways, **inclusive**
- L1 dTLB reach: 256 KB
- L2 TLB reach: 6MB

Challenges: Finding Threshold



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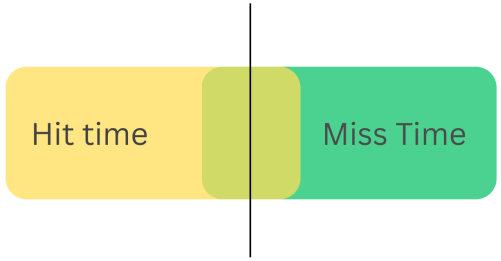

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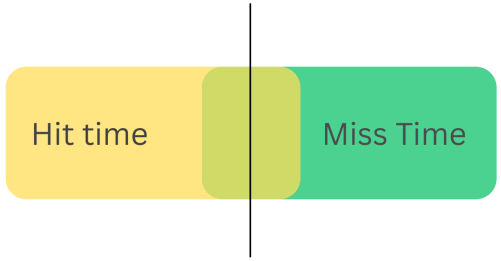

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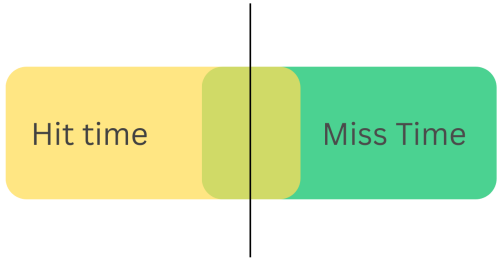

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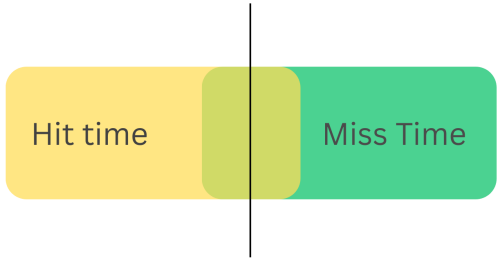

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- Context switches create outliers in data.

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- Typical LLC Hit time = 100 rdtsc cycles, Typical Miss time = 300+ rdtsc cycles
- Our **Threshold = 200** rdtsc cycles
  Hit correctly classified = 99.9964%
  Miss correctly classified = 99.9999%

Challenges: Finding Threshold

- Prefetcher can identify the access pattern and cause false hits.

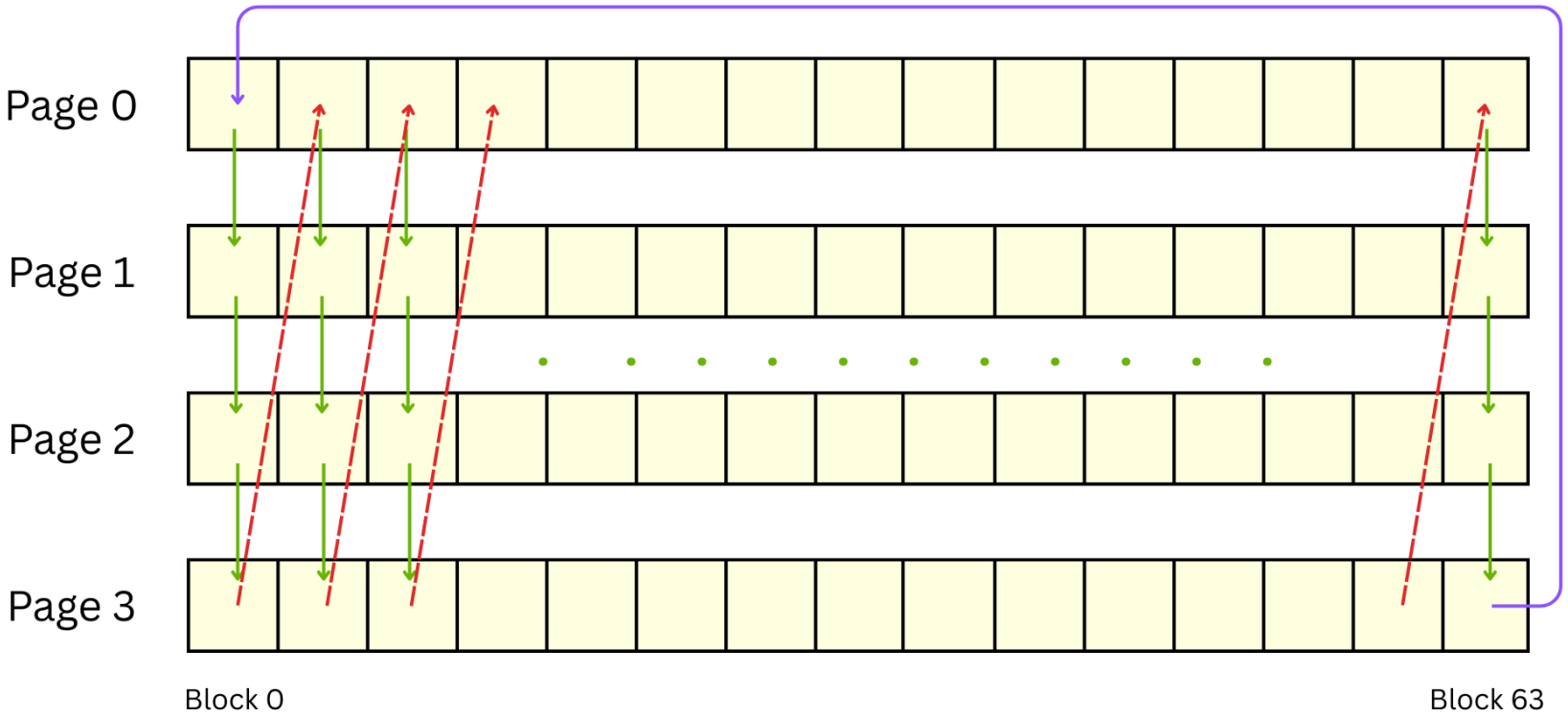

Figure: Access Pattern to fool the prefetcher

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- Need to isolate a relatively quiet set
- **Noise** may cause false misses.
- **Prefetcher** can identify access pattern and prefetch our blocks and cause them to hit when we expect a miss.
- Sender needs to find disturbance set size elements mapping to the same cache set which receiver isolated, from a different process.

Challenges: Synchronising sender and receiver

[3]Source : https://www.tutorialspoint.com/inter_process_communication/inter_process_communication_pipes.htm

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- Sender and receiver need to know when they can do their work, and exactly one should work at any point in time for all steps.

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- Sender and receiver need to know when they can do their work, and exactly one should work at any point in time for all steps.
- We have used pipes between sender and receiver to synchronize them.

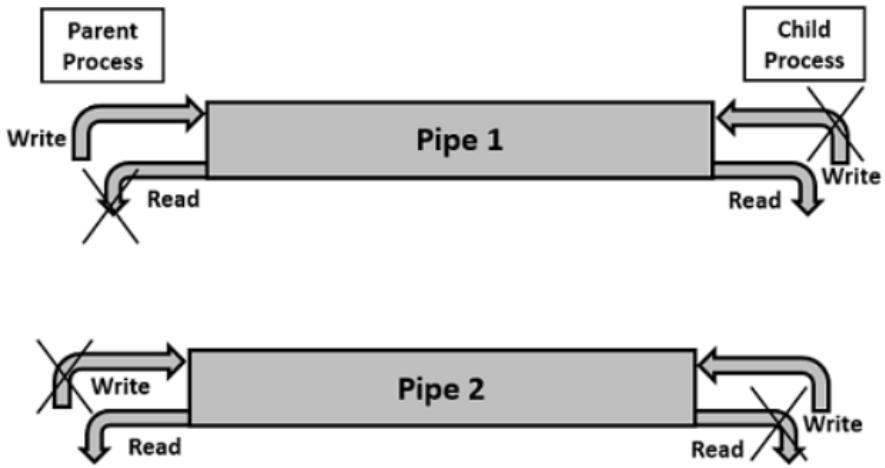

Figure: Inter Process Communication Pipe [3]

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Overview of Steps

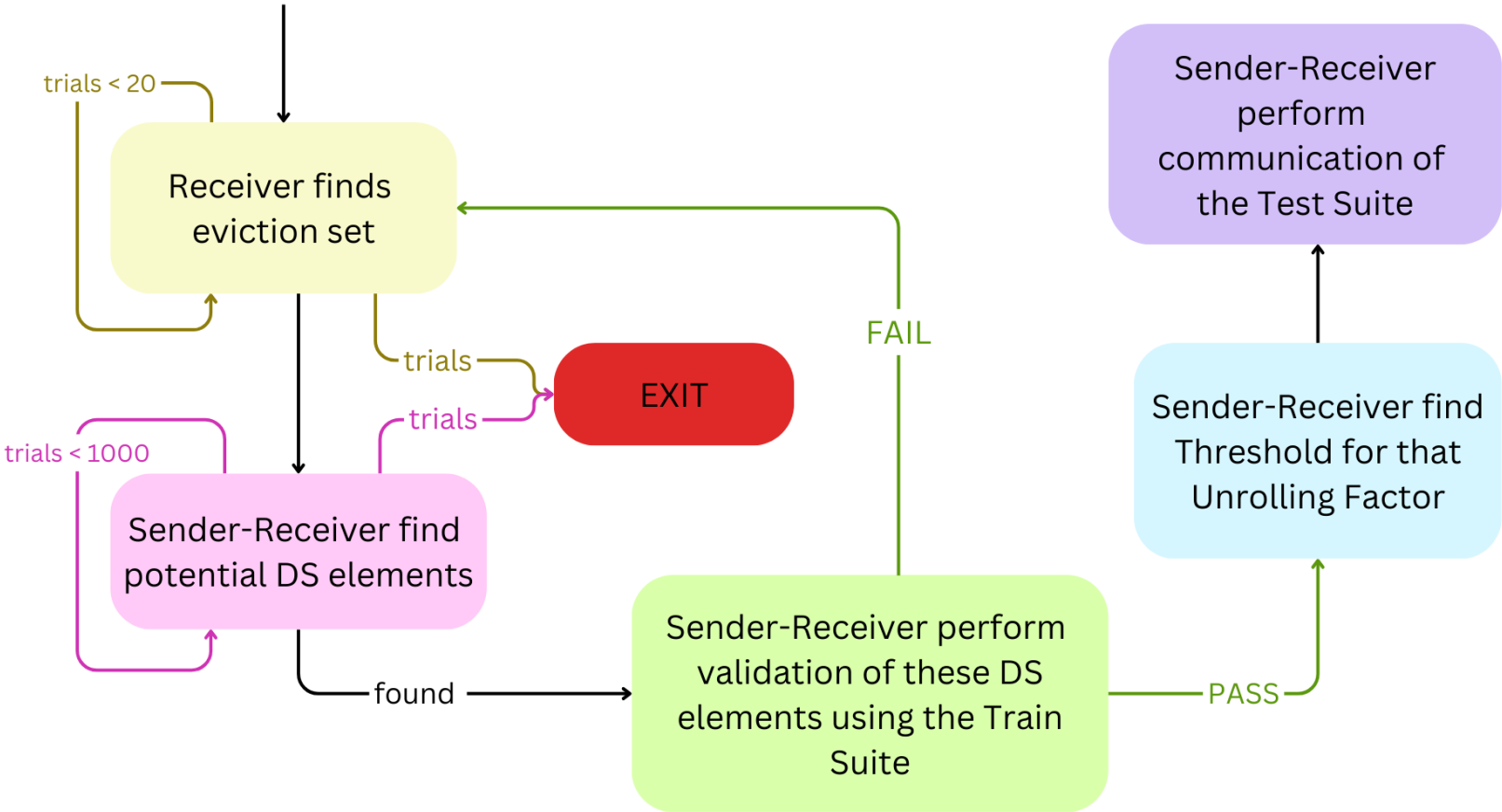
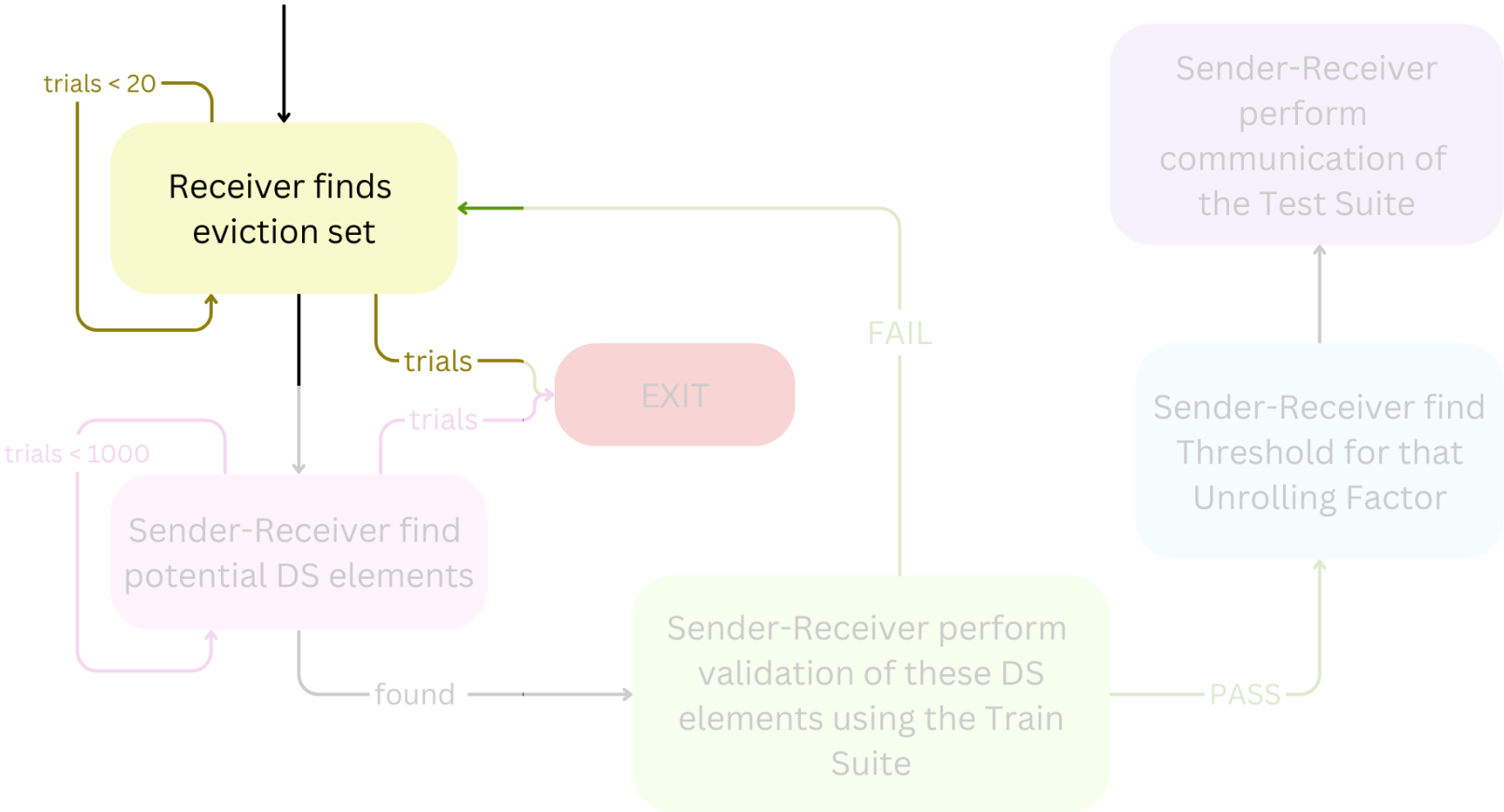


Figure: Steps for establishing Covert Channel

Finding an eviction set for the receiver



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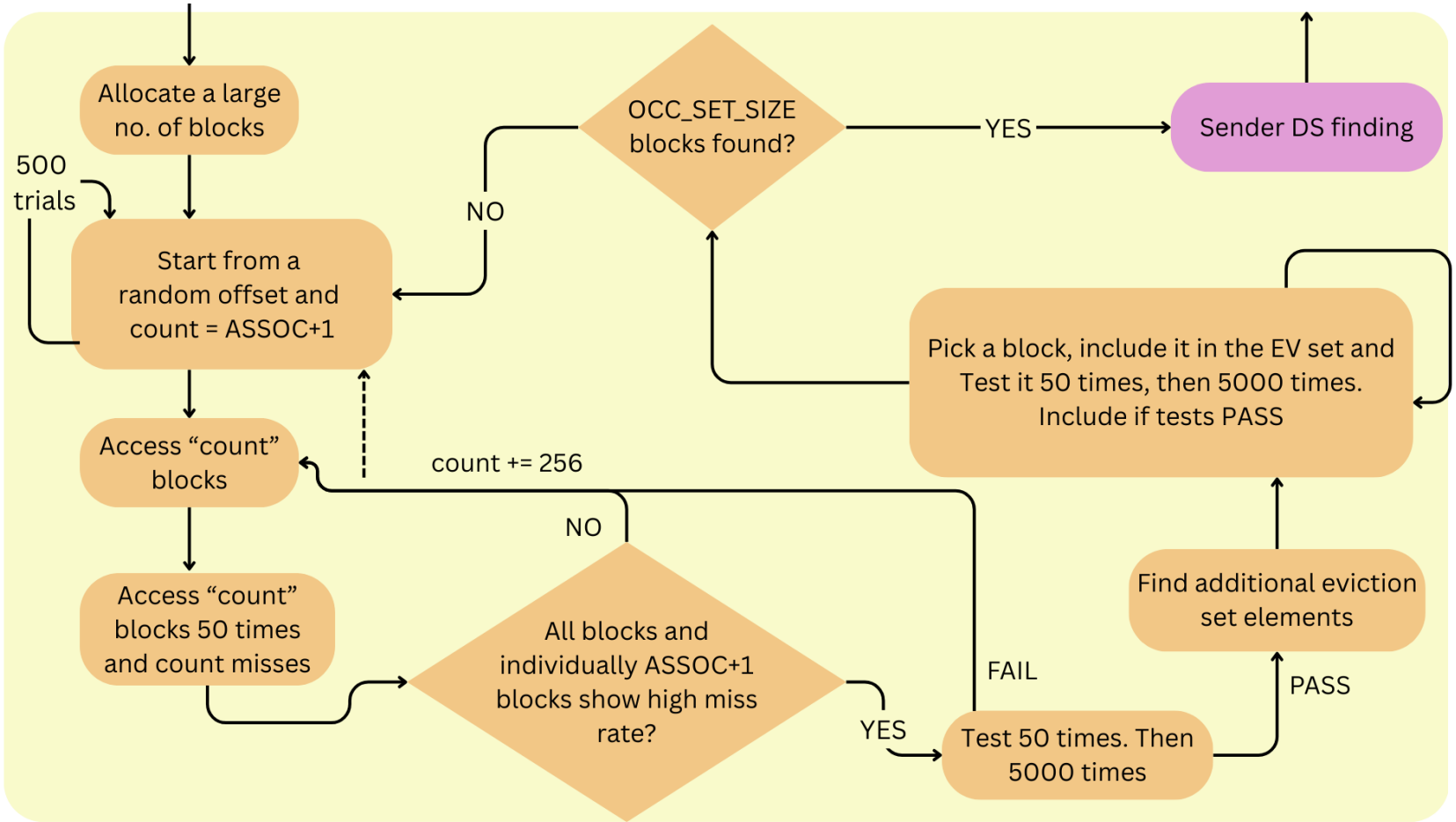

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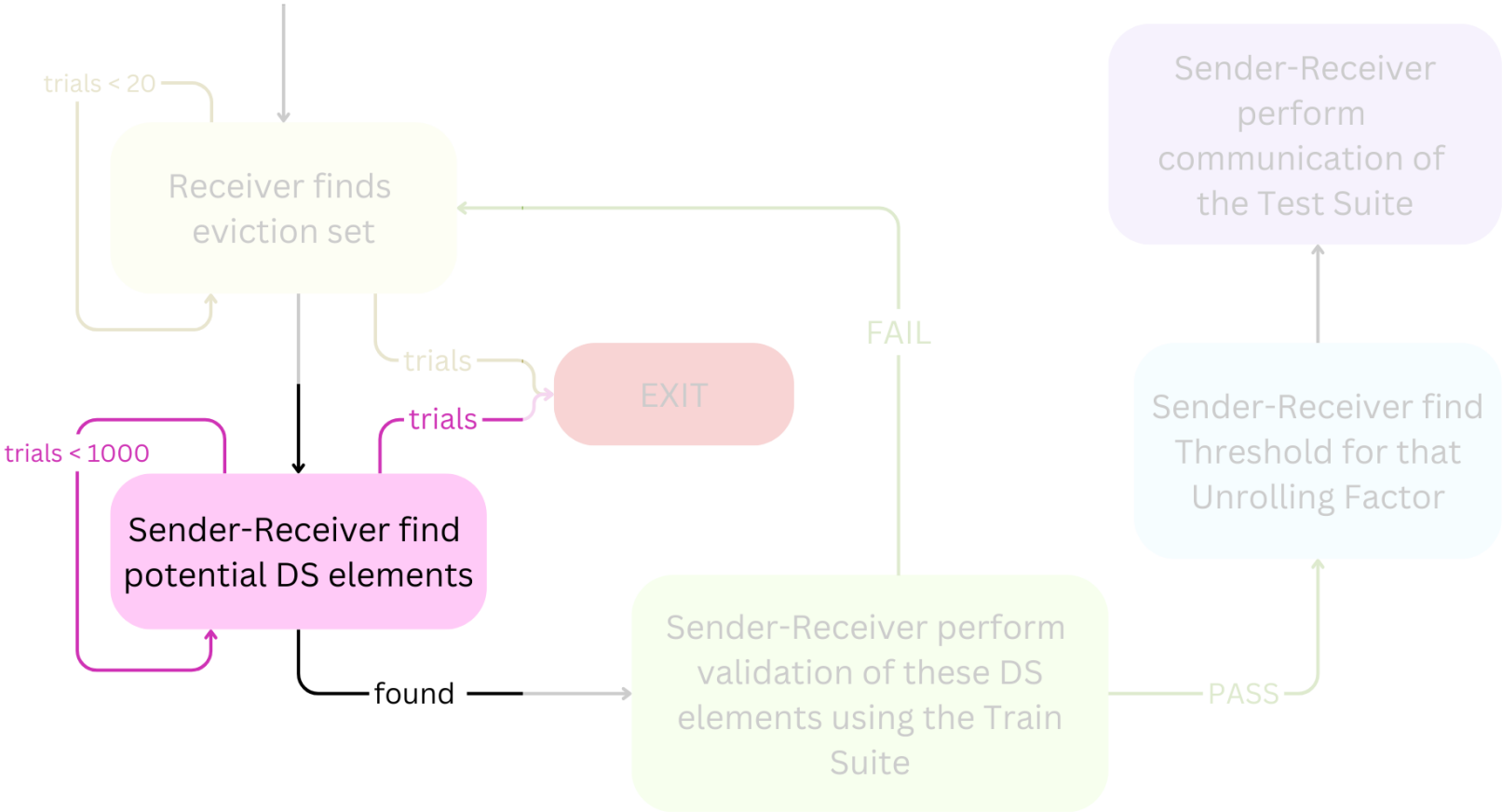
Finding disturbance set for the sender



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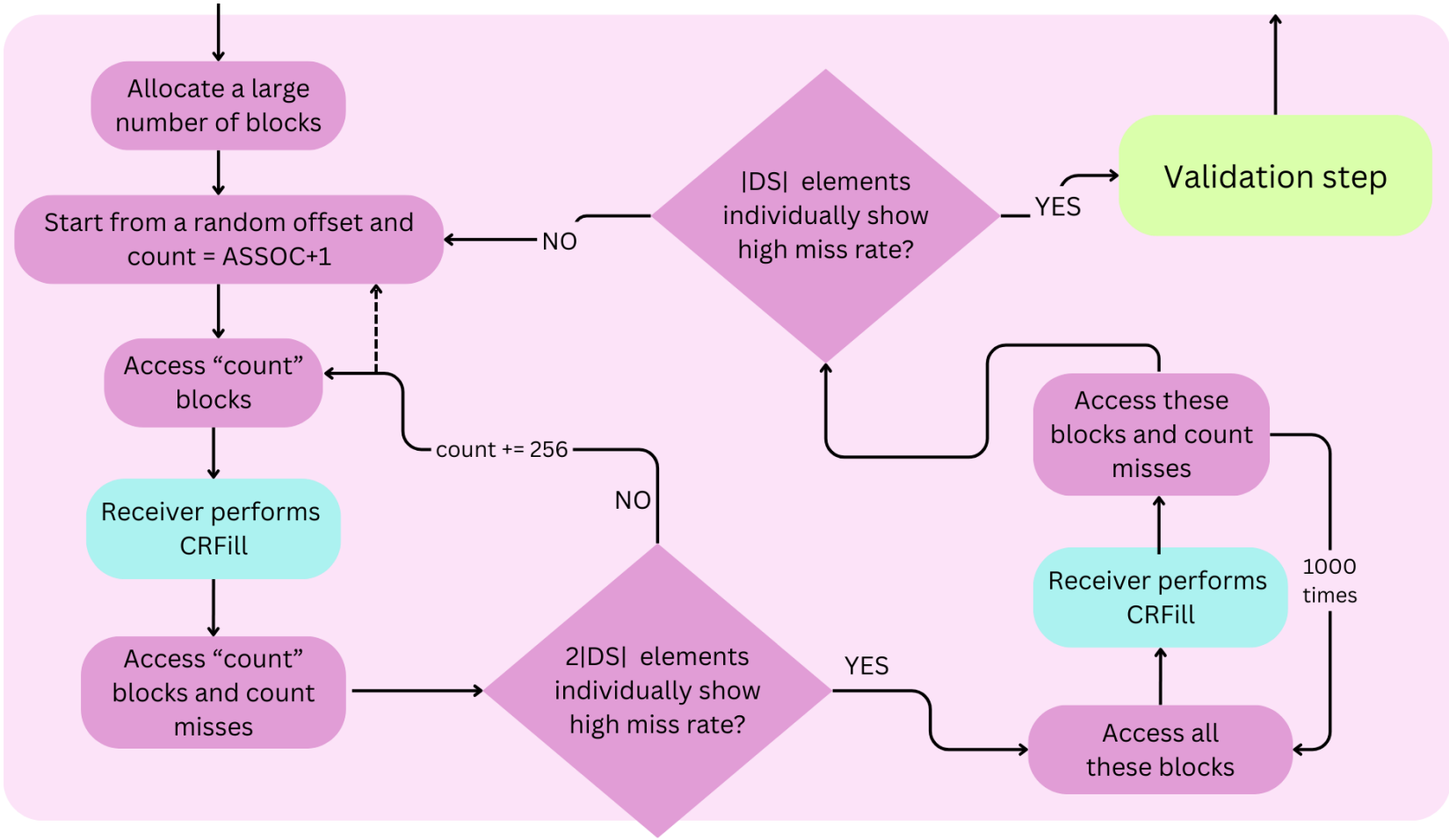

Figure: DS for sender

Validation of disturbance set using Train_suite

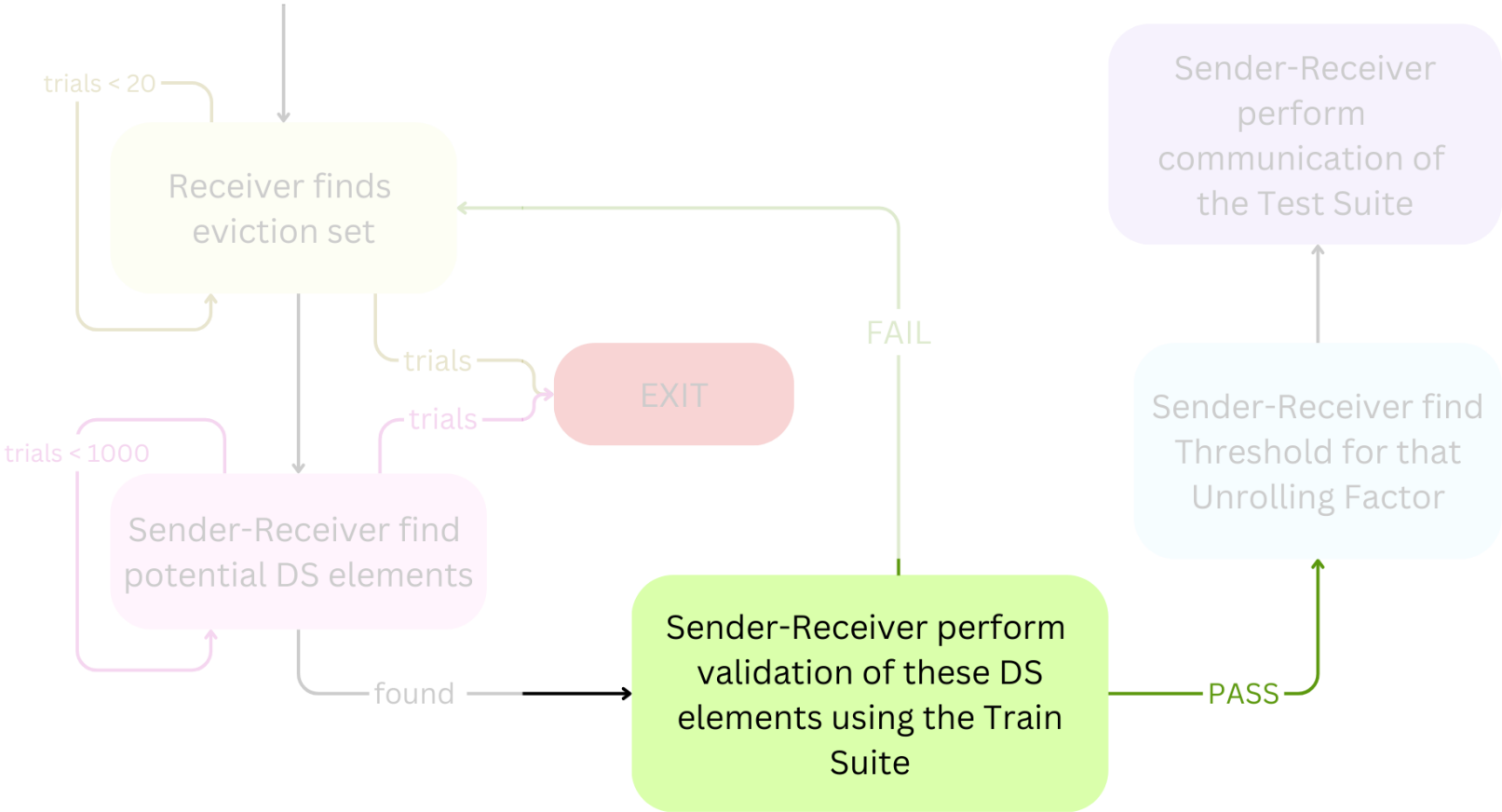

Figure: Validation step

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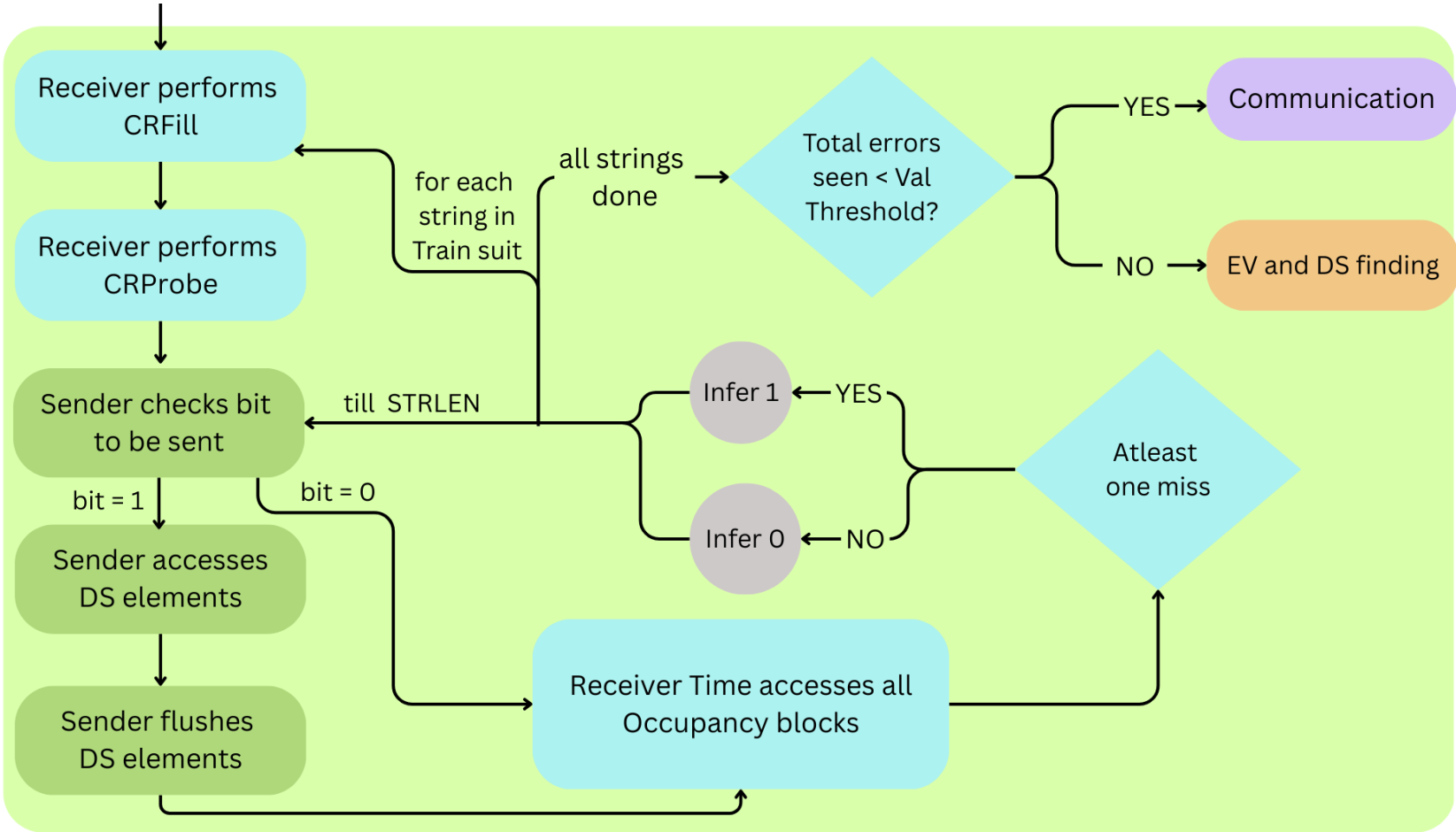

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Finding Unrolling Factor Threshold for receiver

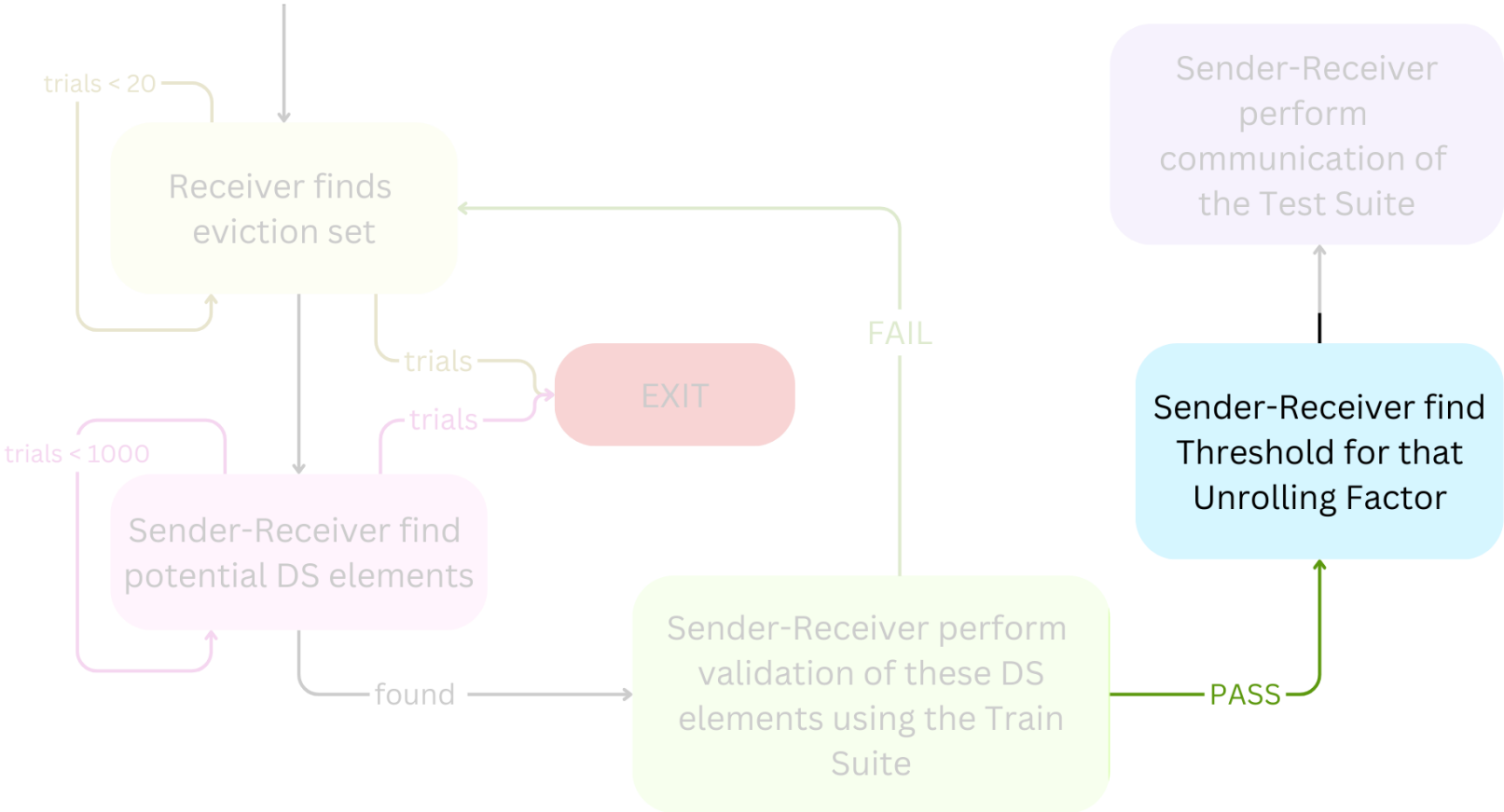

Figure: UF Threshold

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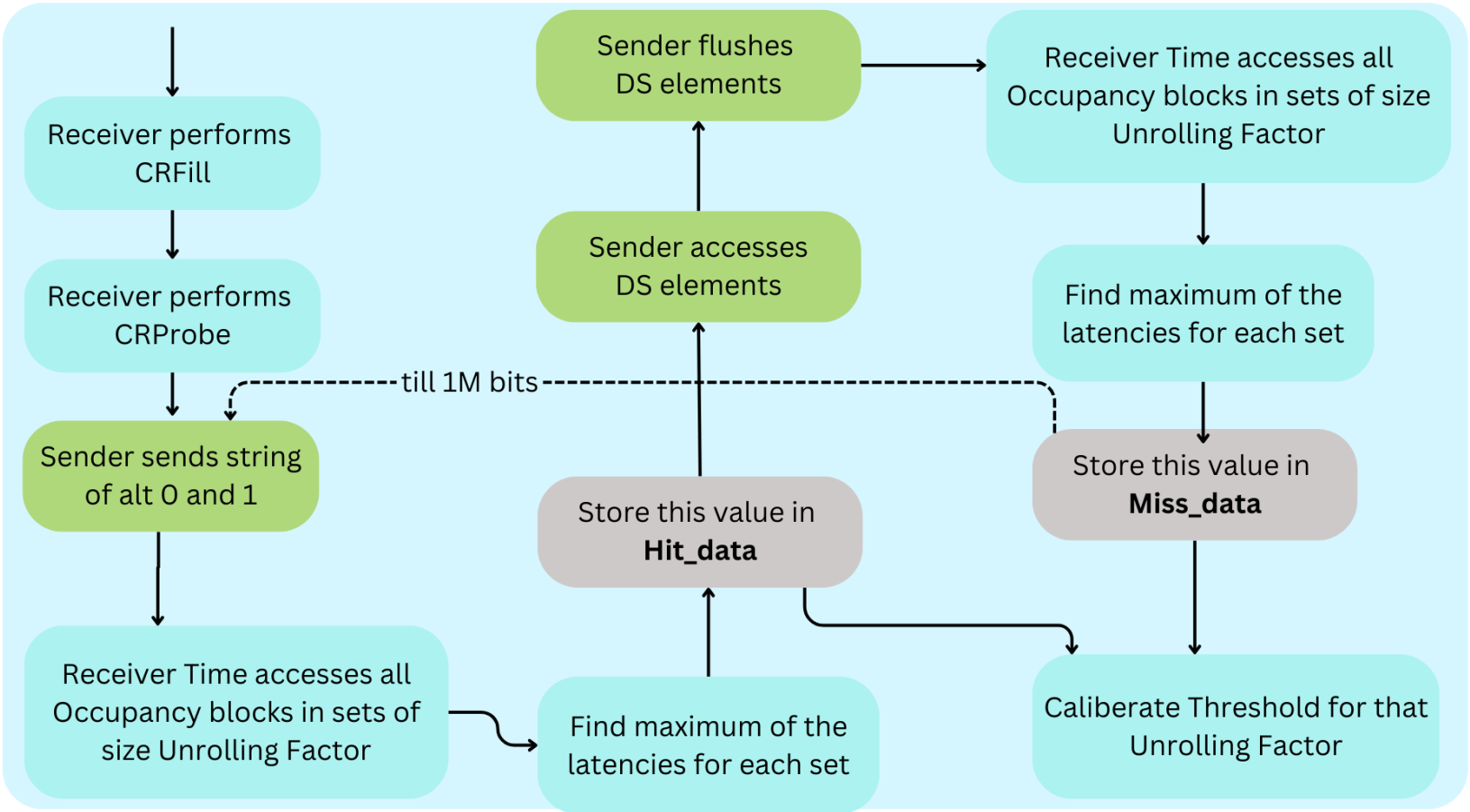

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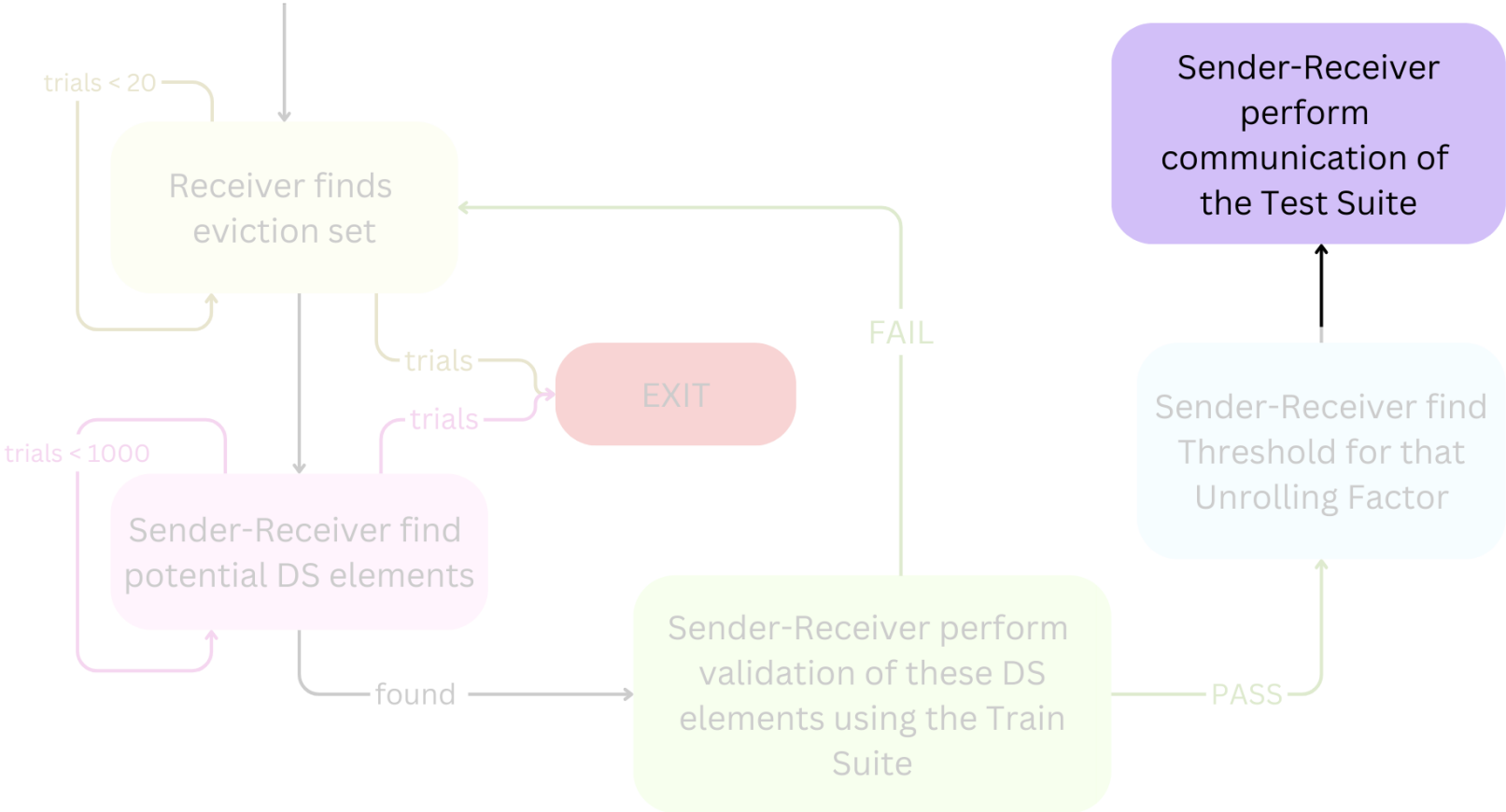

Figure: Communication

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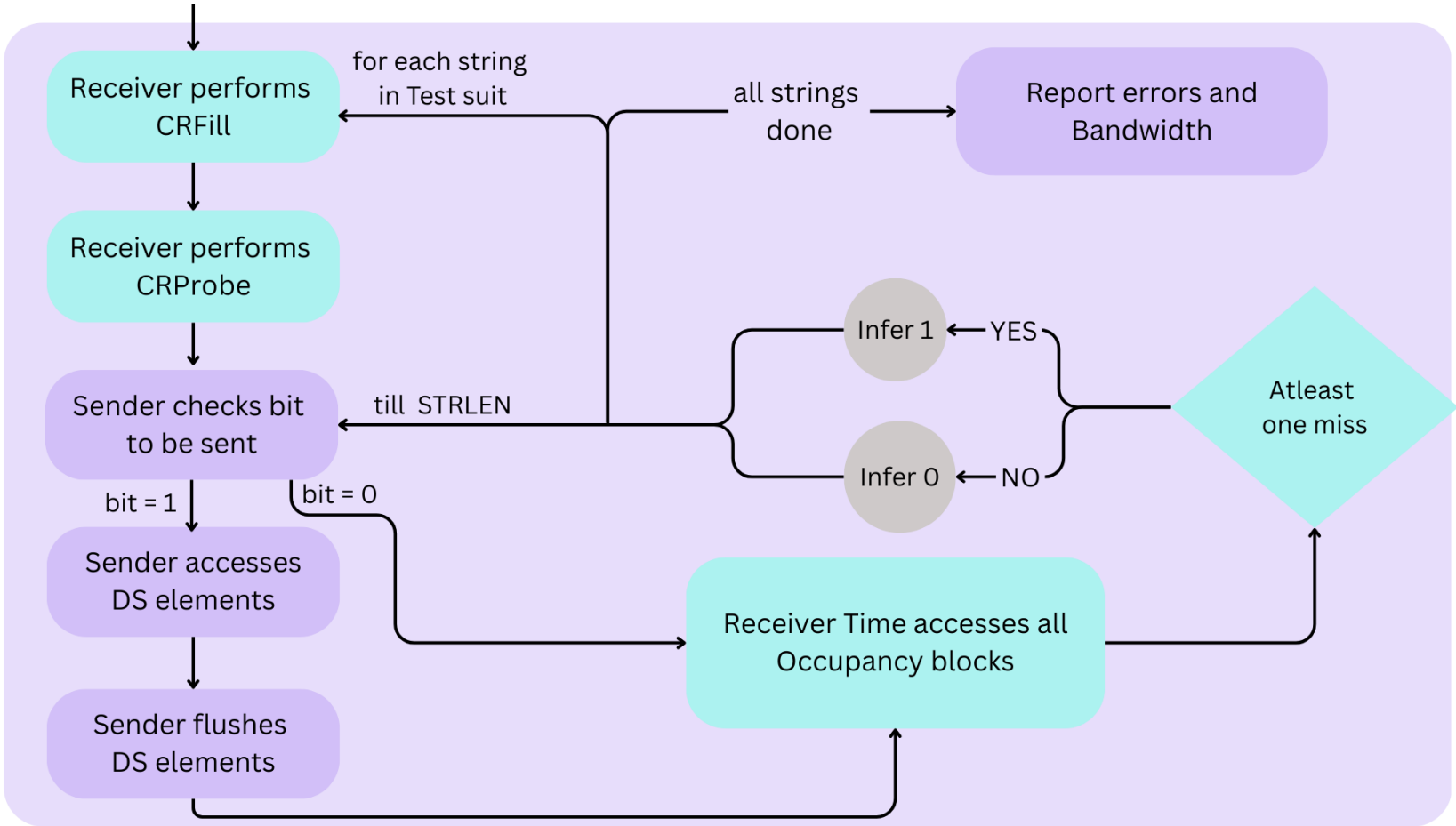

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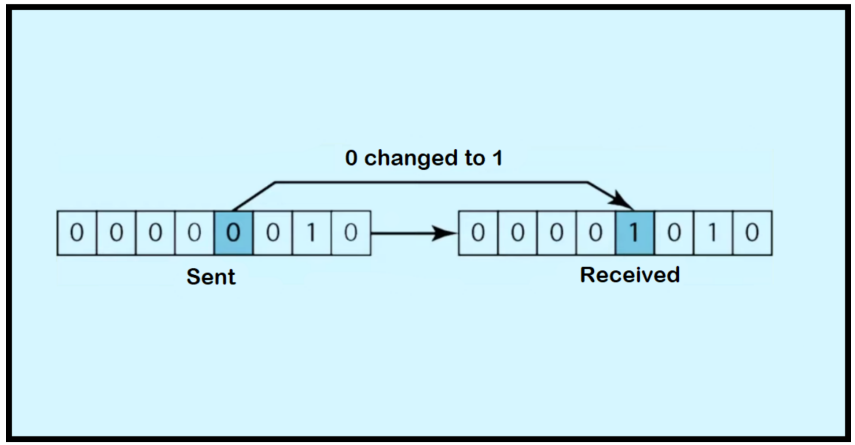

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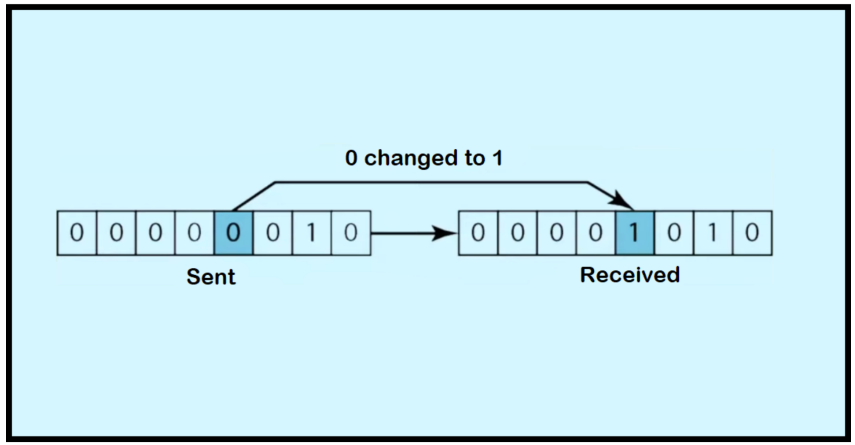

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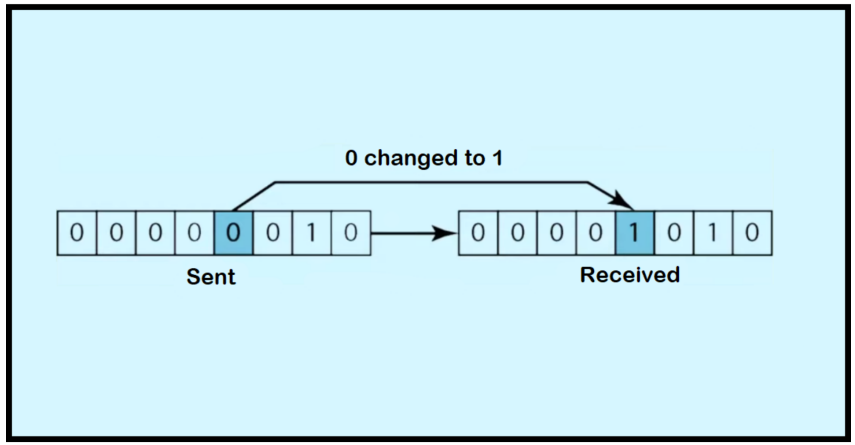

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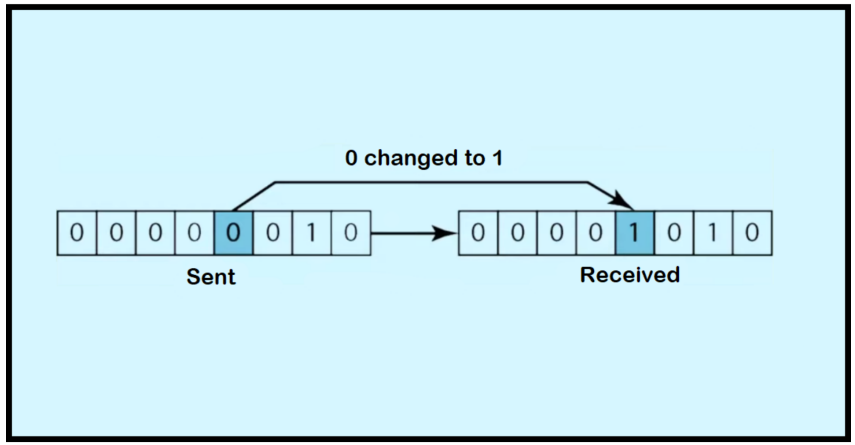

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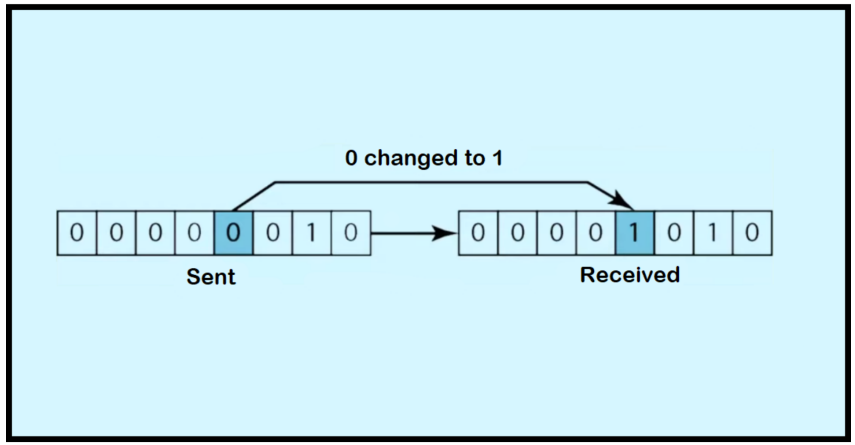

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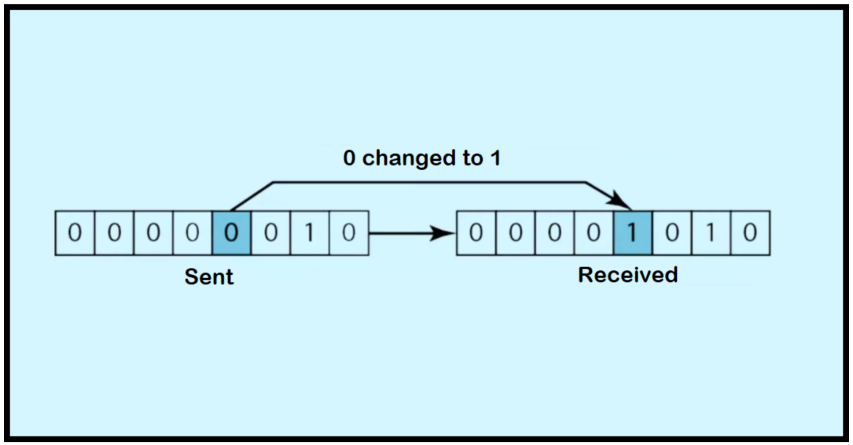

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- OS noise - context switches, syscalls, page faults
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- Larger overlap in hit and miss latencies

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- Don't use syscalls (for eg., printf) during communication.

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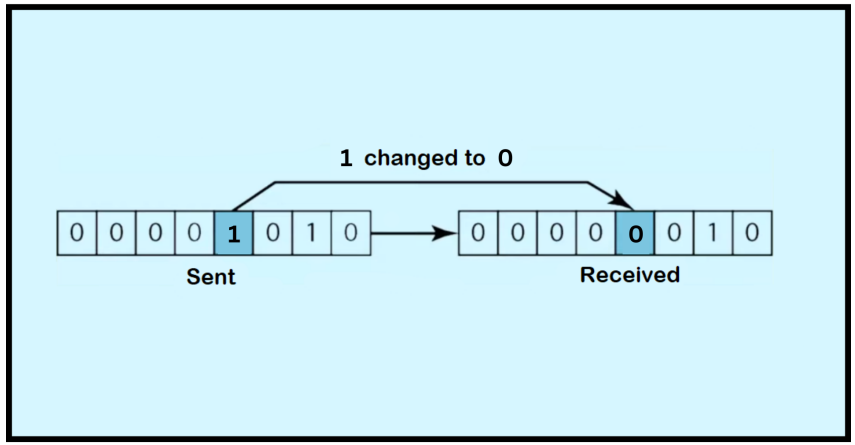

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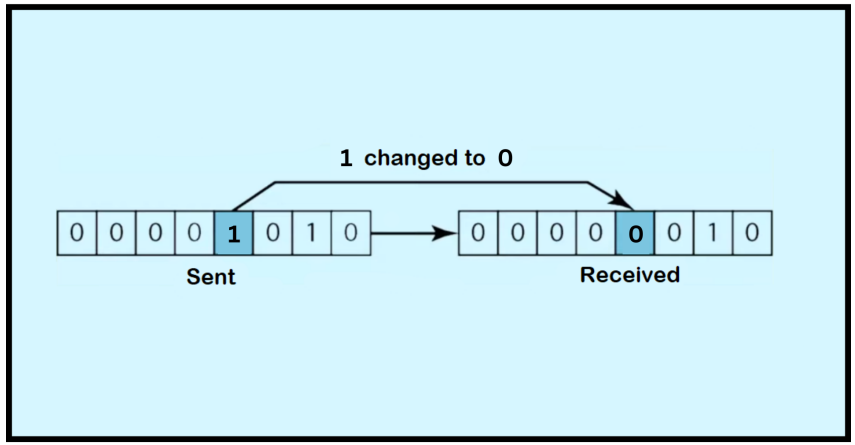

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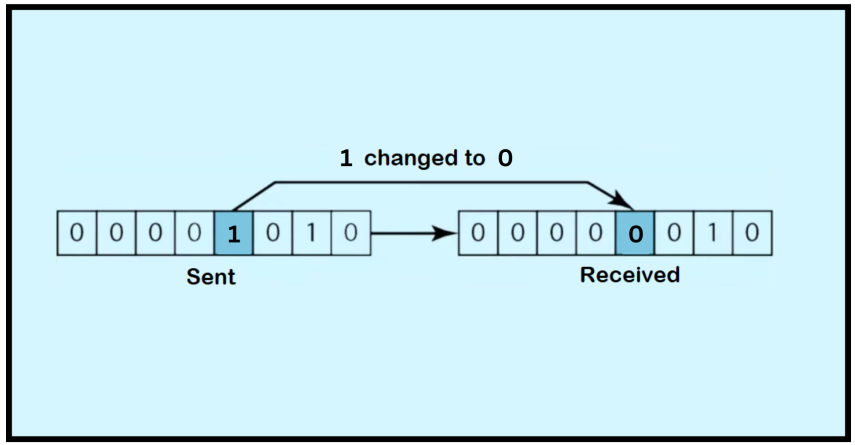

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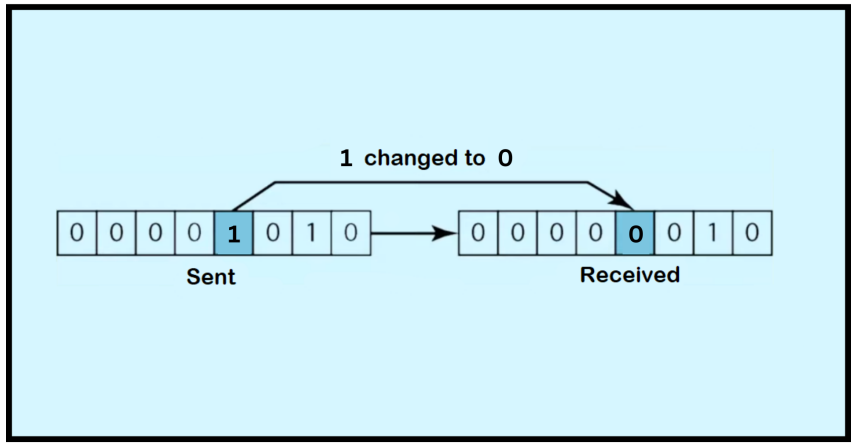

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- Poor occupancy achieved by receiver
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- If the eviction set and DS elements map to the same set in the LLC, any disturbance created by the sender will indeed evict atleast one of the receiver's blocks.
- The overlap in hit and miss latencies should be the main culprit.

Results & Observations

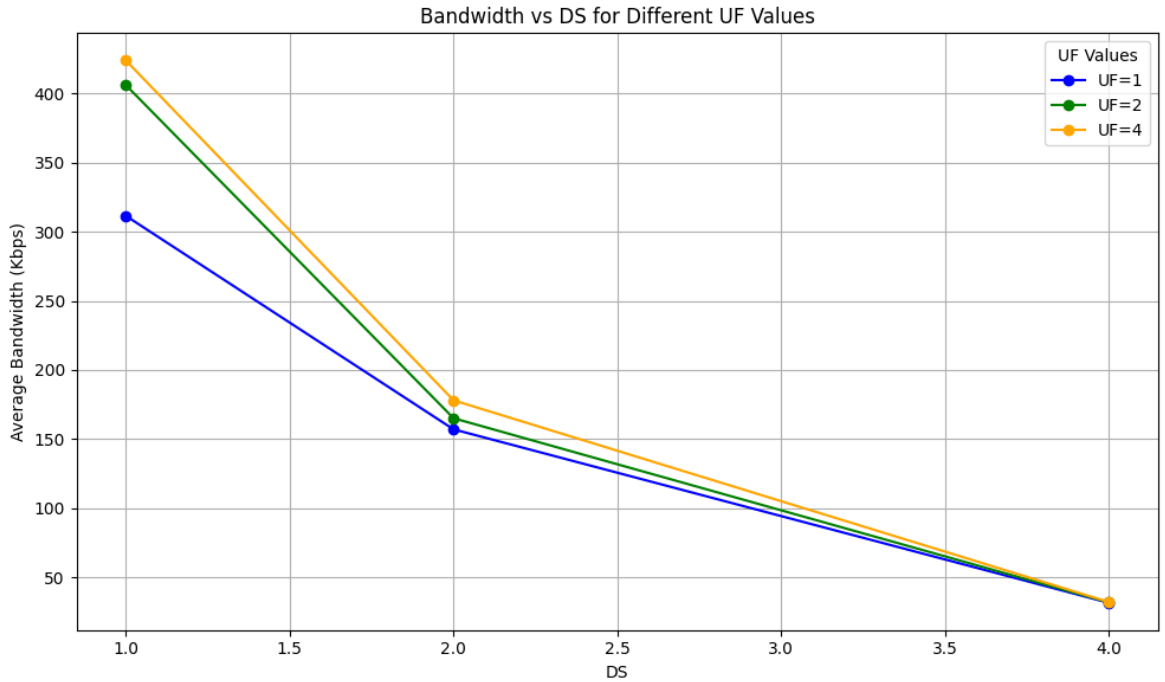

Figure: Bandwidth vs DS

- Bandwidth decreases on increasing DS (significantly)
- Bandwidth increases on increasing unrolling factor (slightly)

Results & Observations

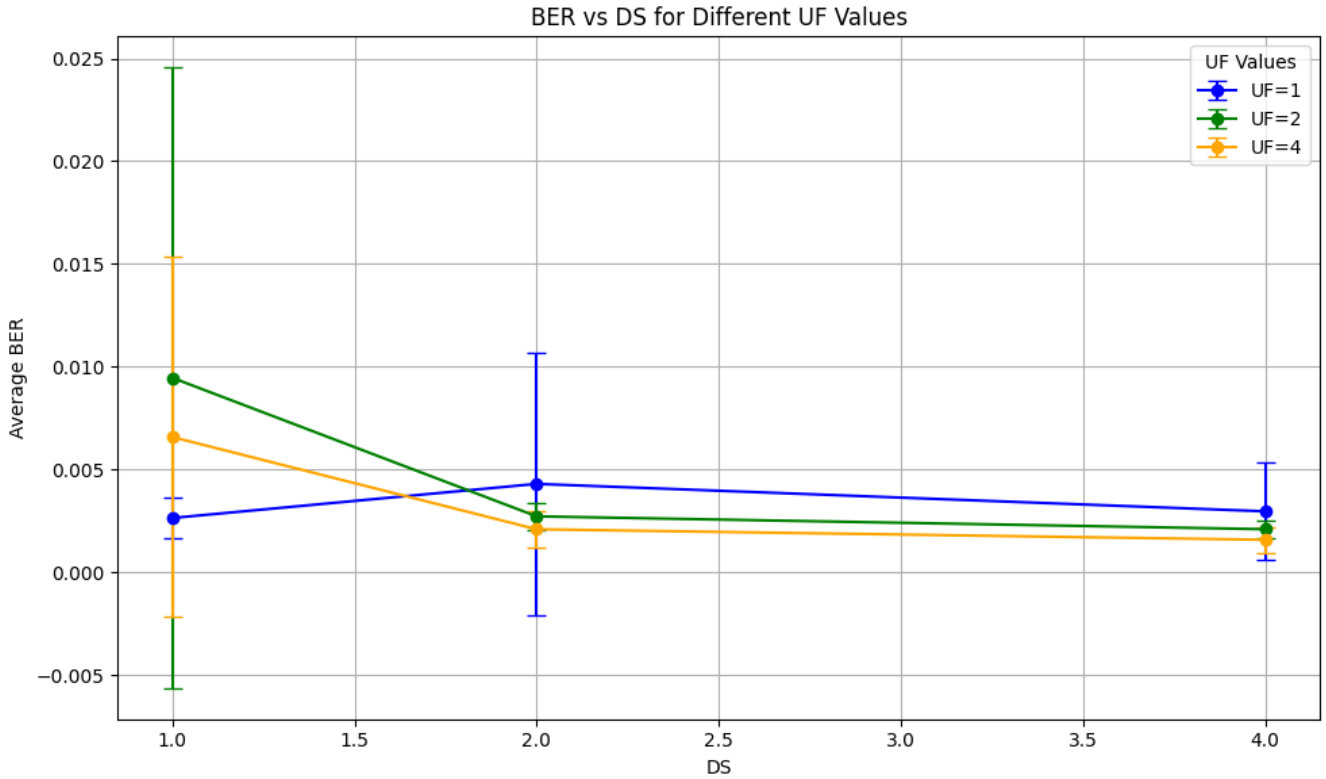

Figure: BER vs DS

- BER decreases on increasing disturbance set size

Results & Observations

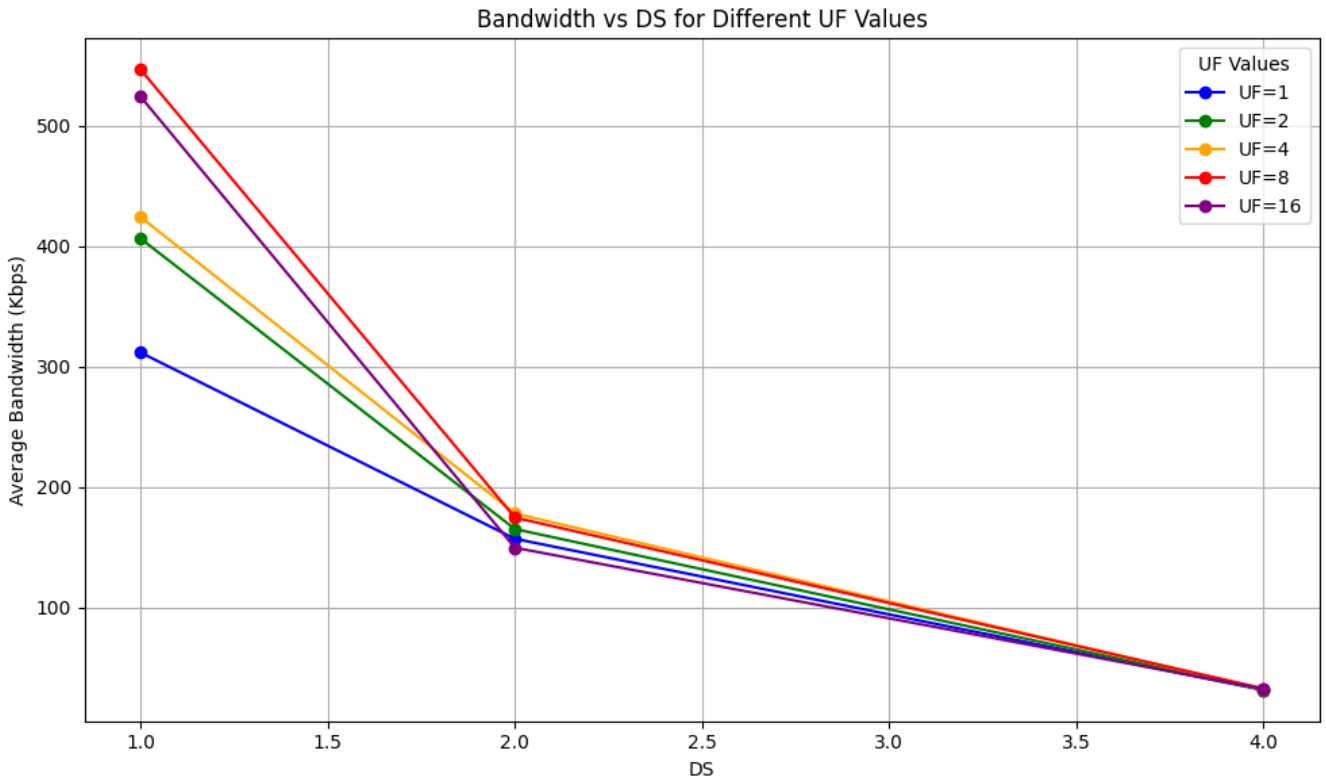

Figure: Bandwidth vs DS

- UF = 8, 16 with DS = 1 have best bandwidths out of all configurations

Results & Observations

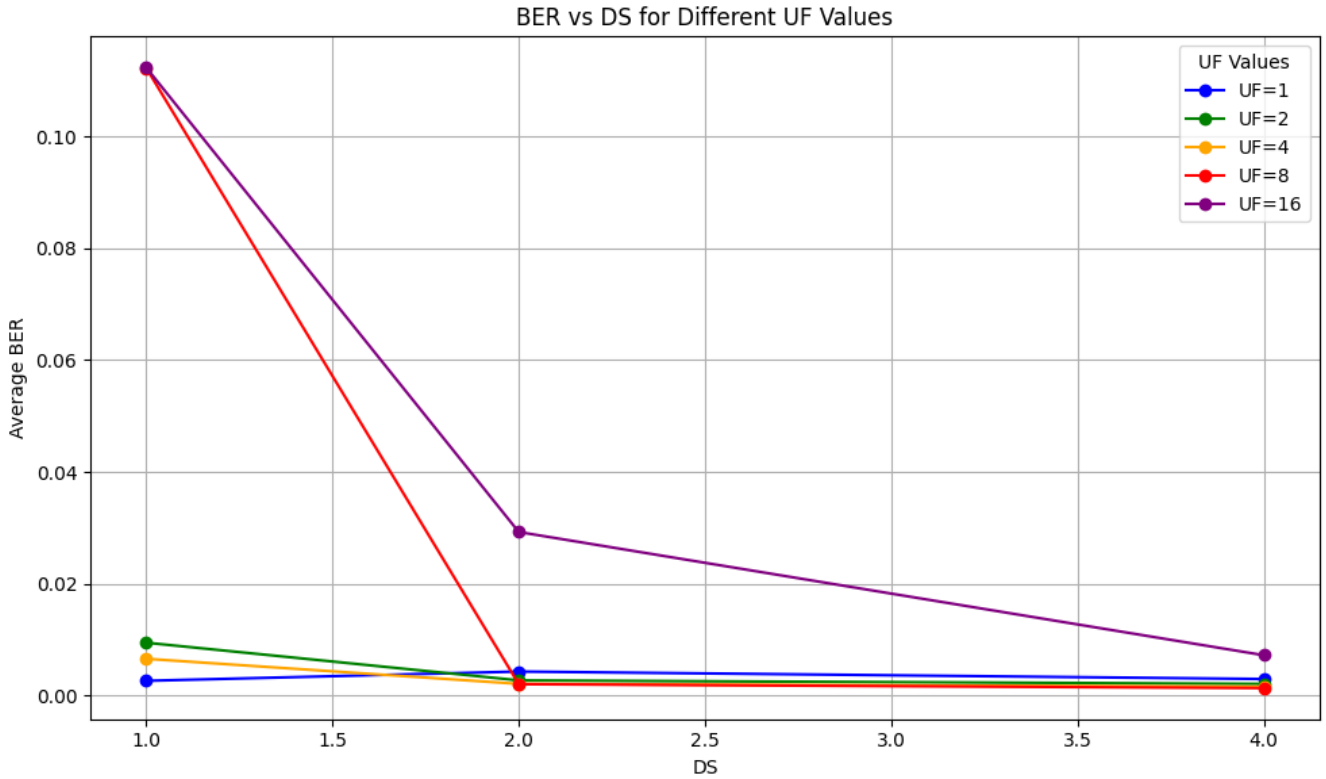

Figure: BER vs DS

- But, the BER is too large!

Future Improvements

- Synchronization of sender and receiver without pipes.
- Ideally there should not be any shared memory between processes.
- Synchronization can be done using the worst case timings for each step.

Thank you for Attending!

Questions and Answers!

Results & Observations

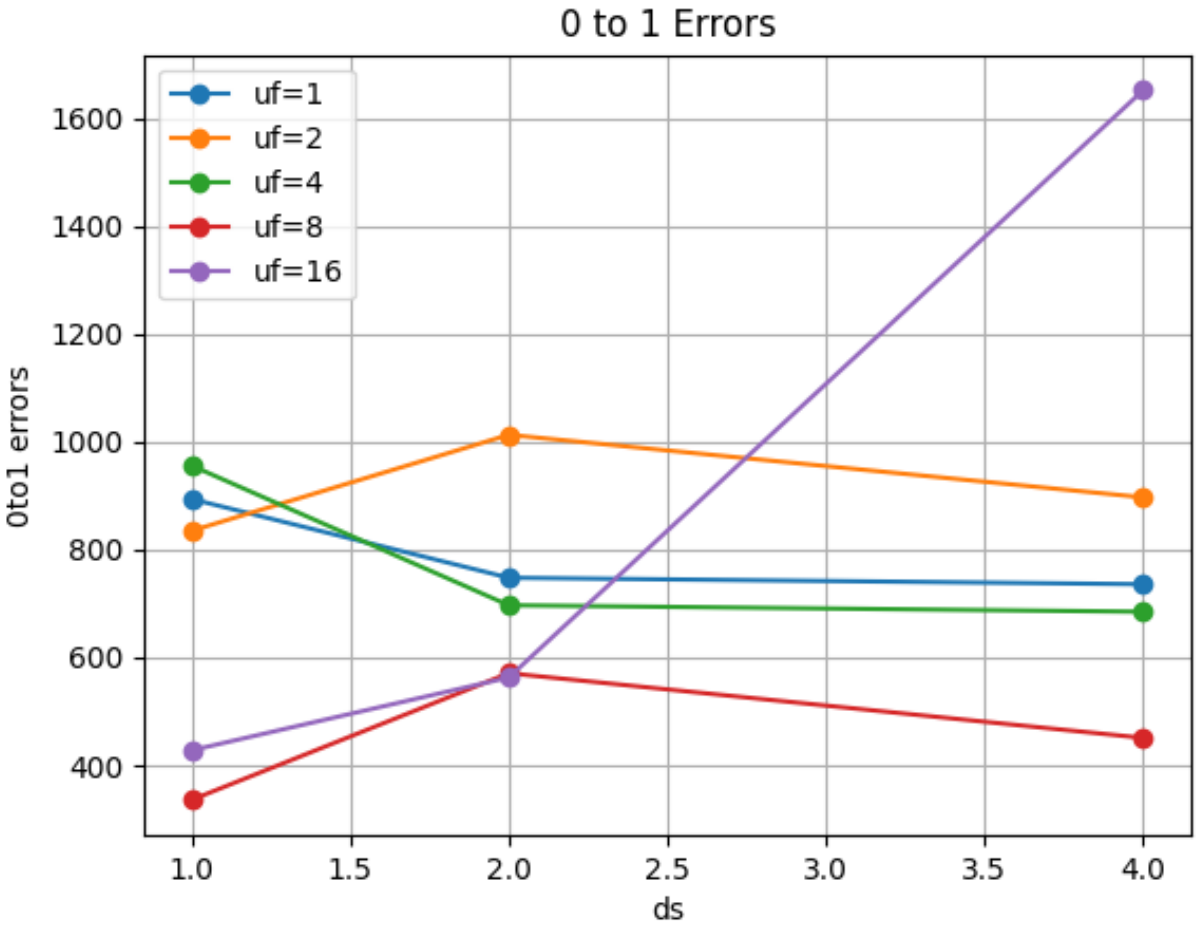

Figure: BER vs DS (0 to 1 errors)

Results & Observations

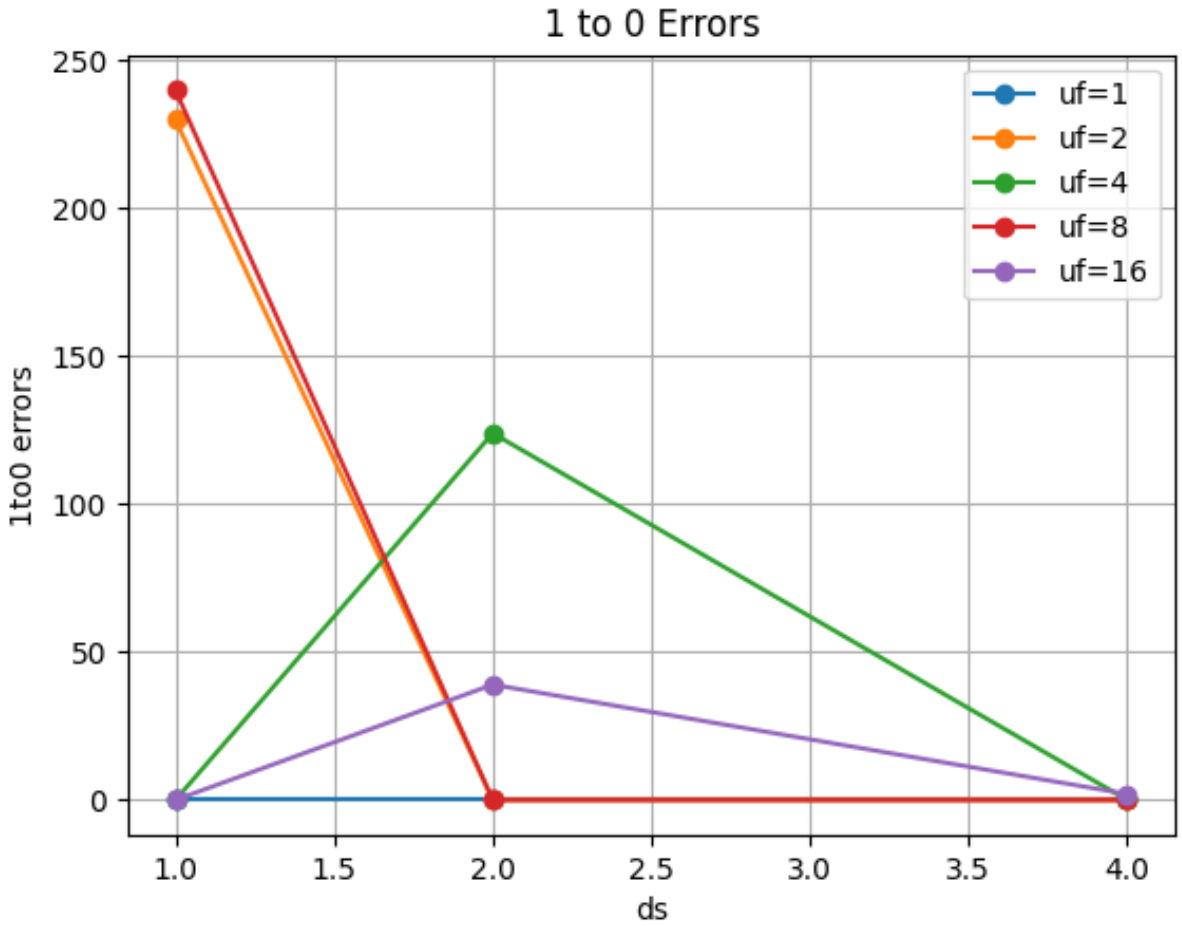

Figure: BER vs DS (1 to 0 errors)

Results & Observations

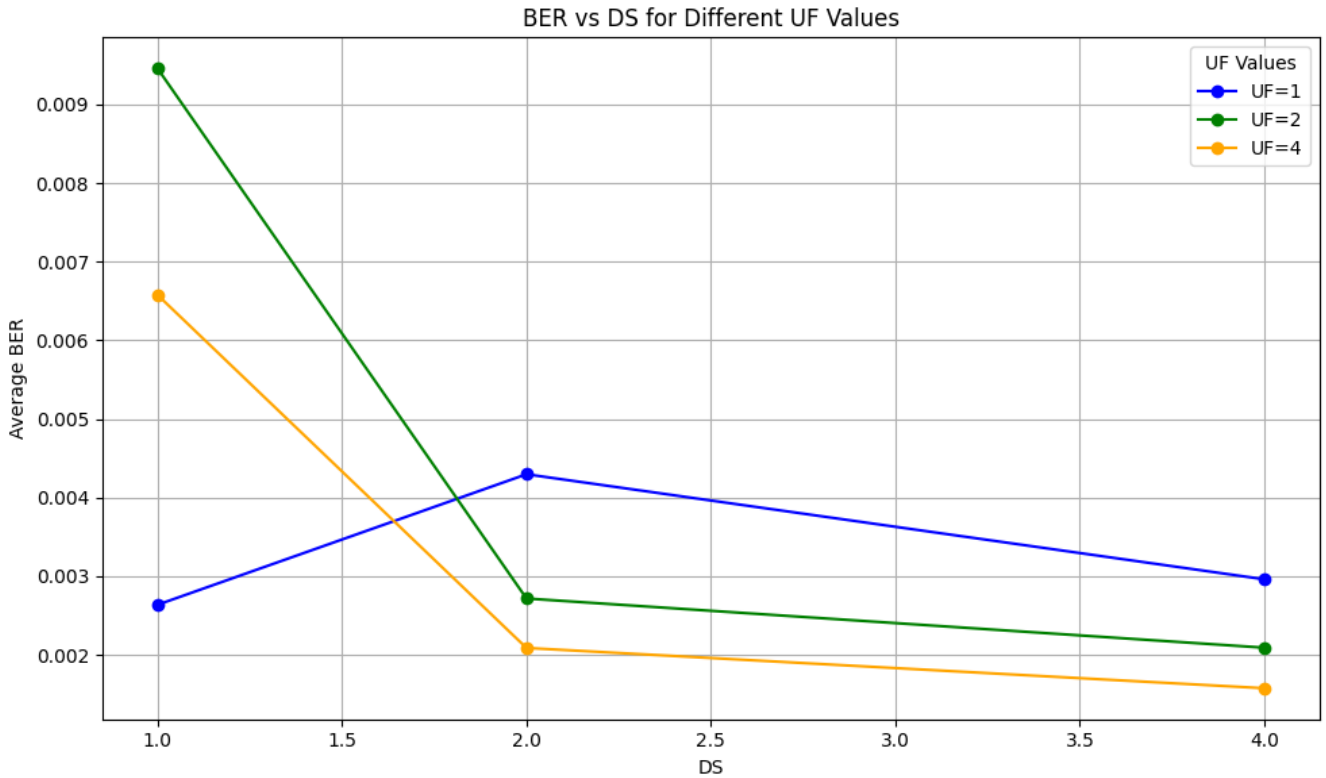

Figure: BER vs DS

Results & Observations

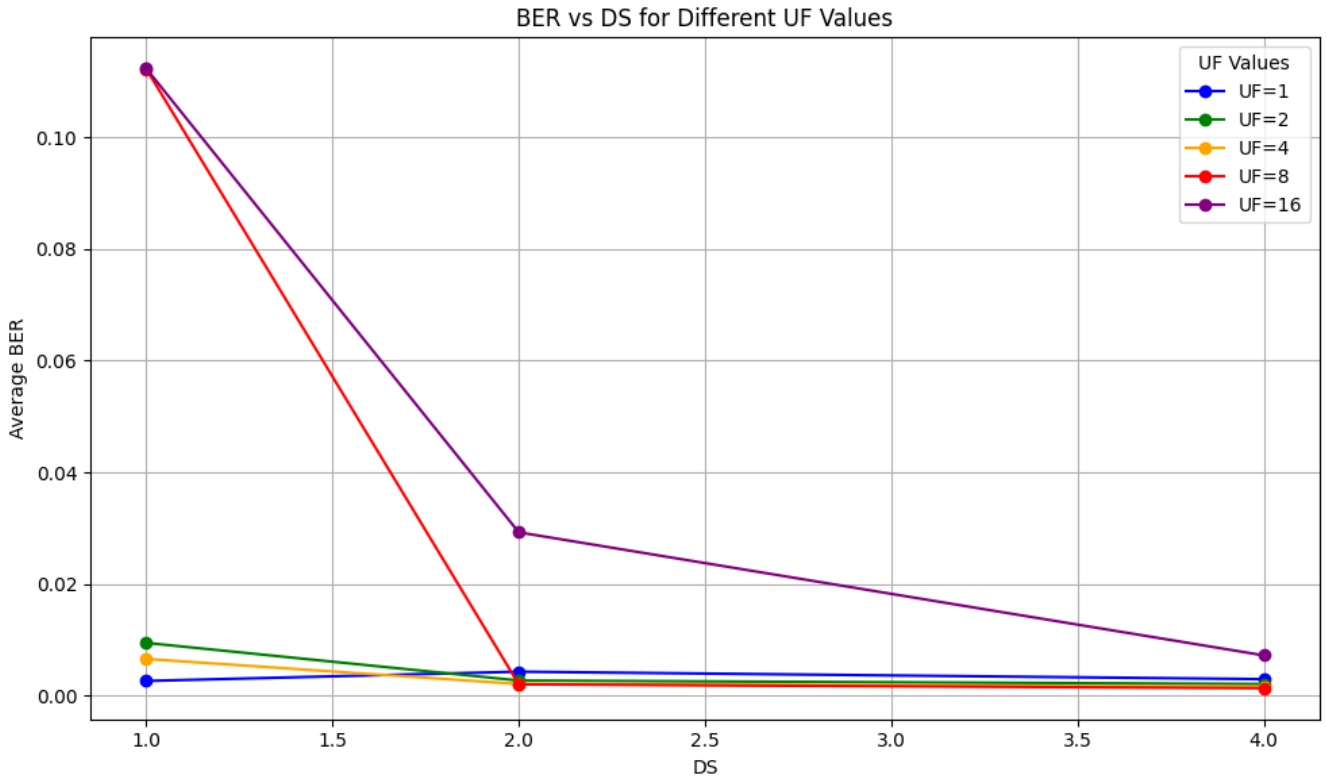

Figure: BER vs DS

Results & Observations

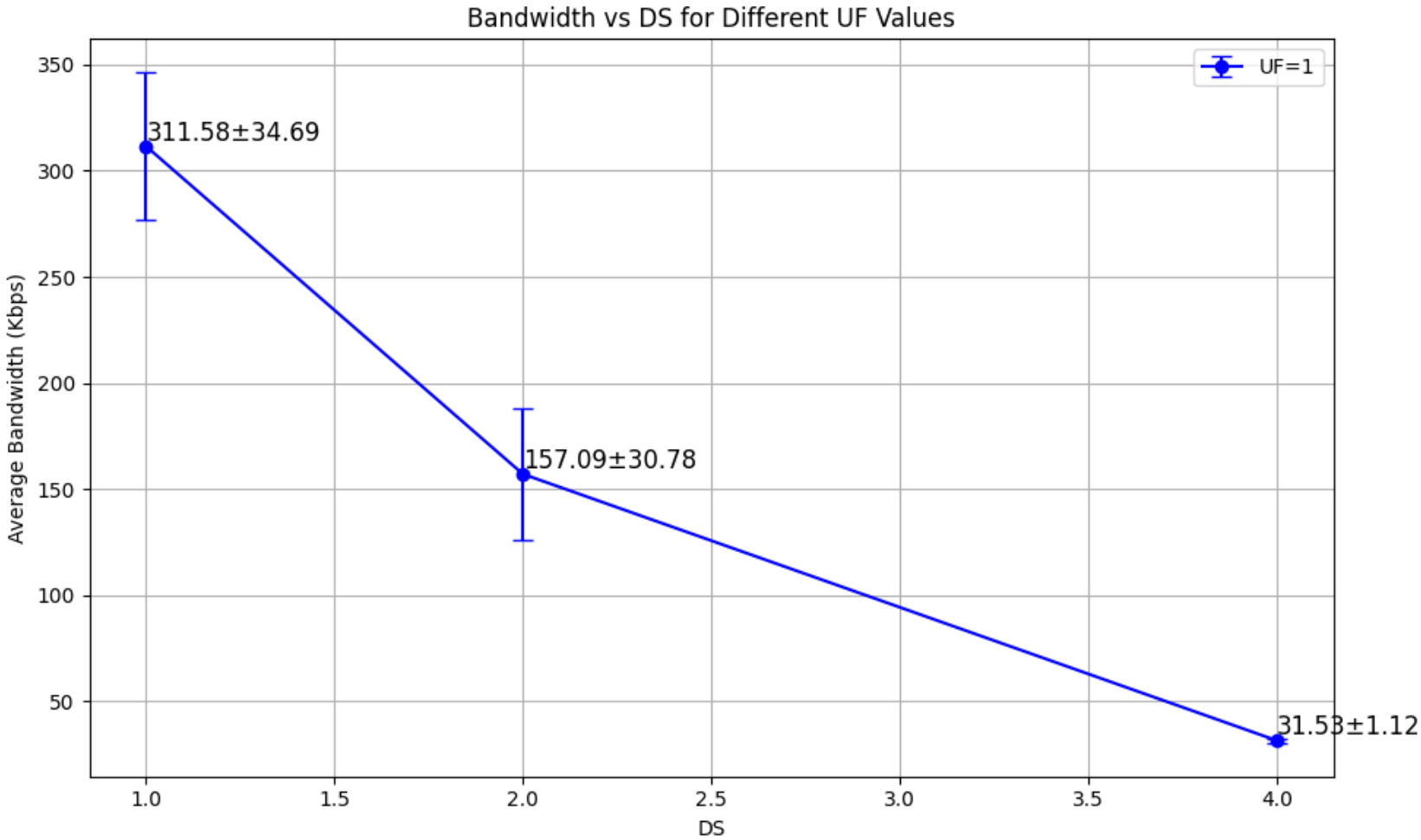

Figure: BW vs DS

Results & Observations

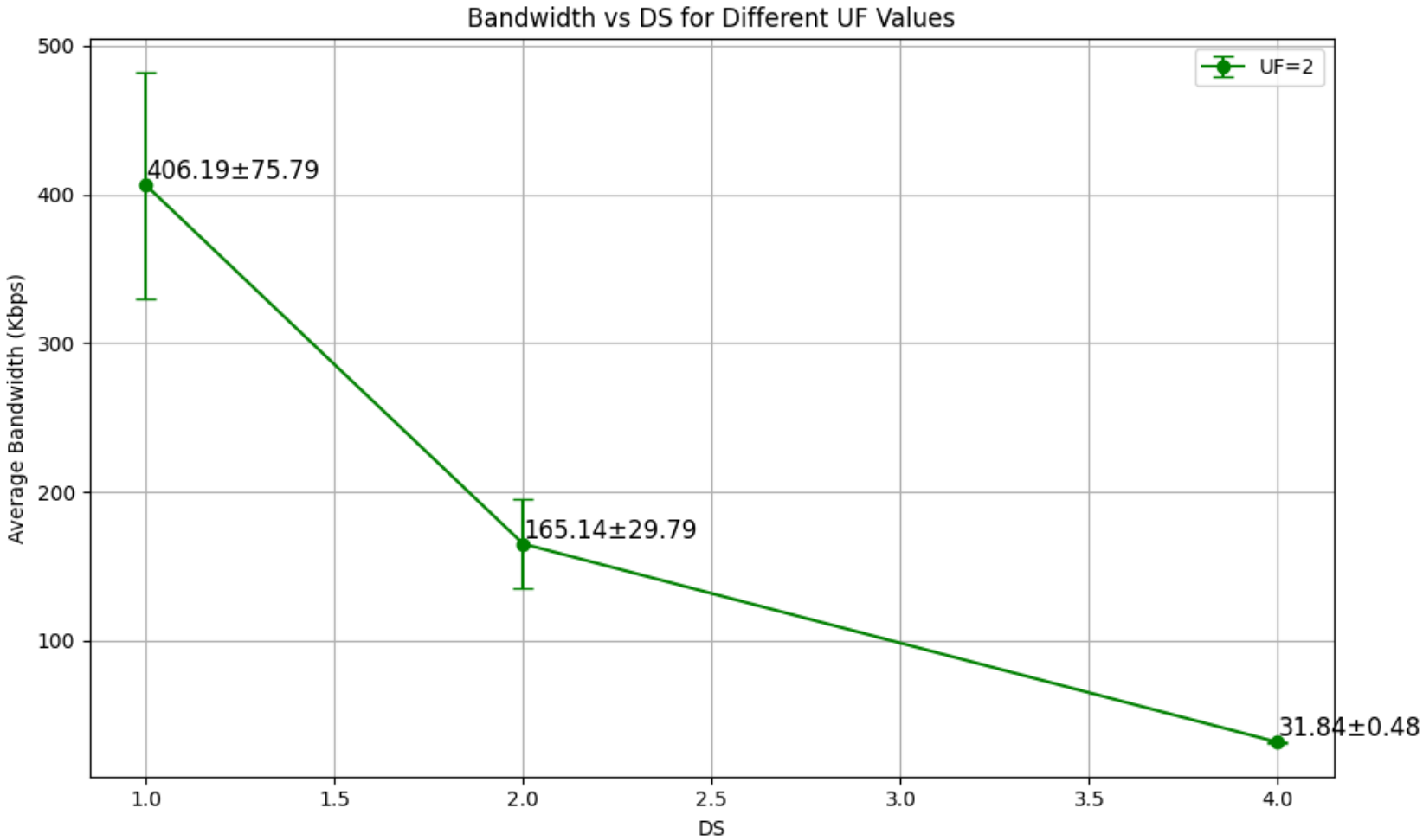

Figure: BW vs DS

Results & Observations

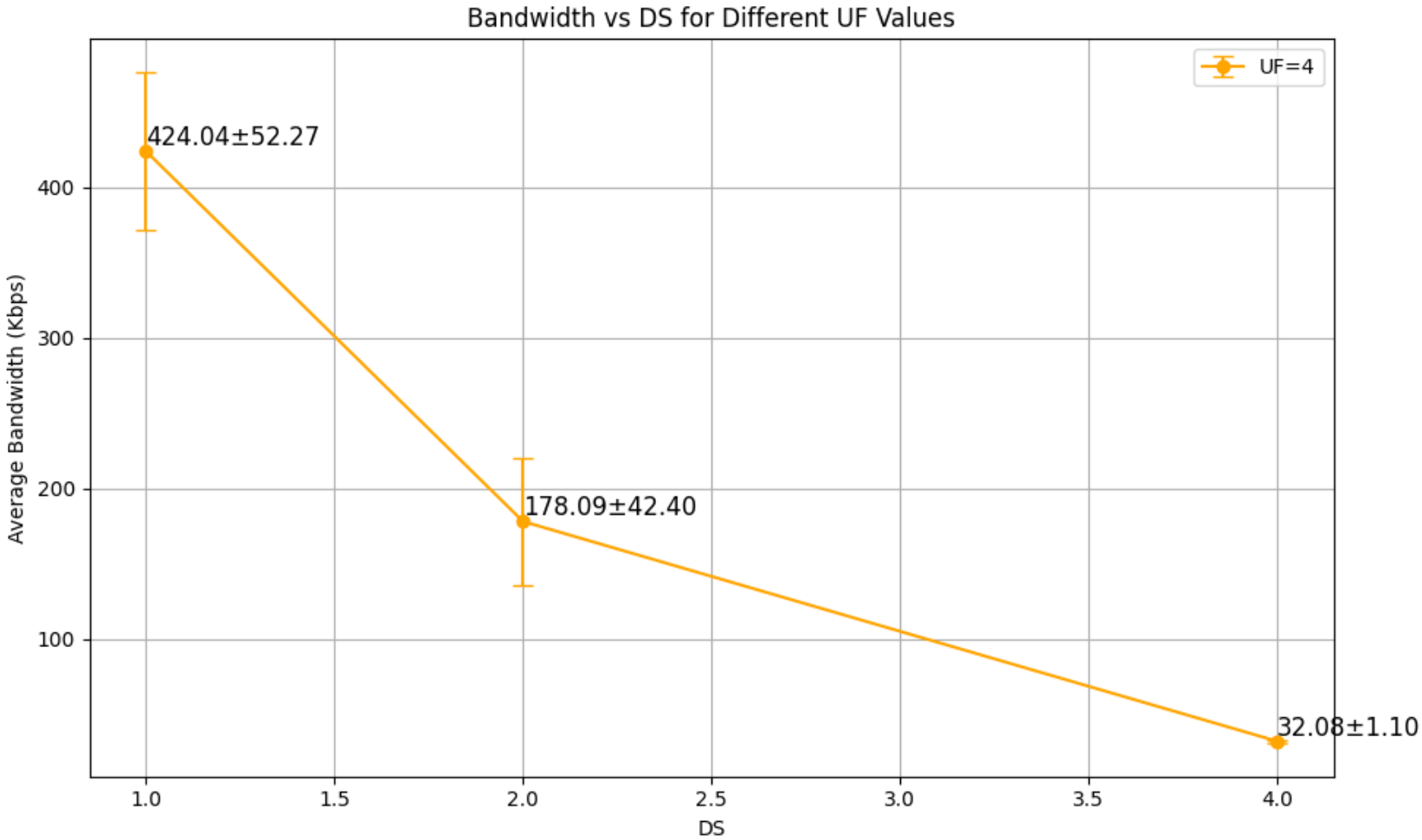

Figure: BW vs DS

Results & Observations

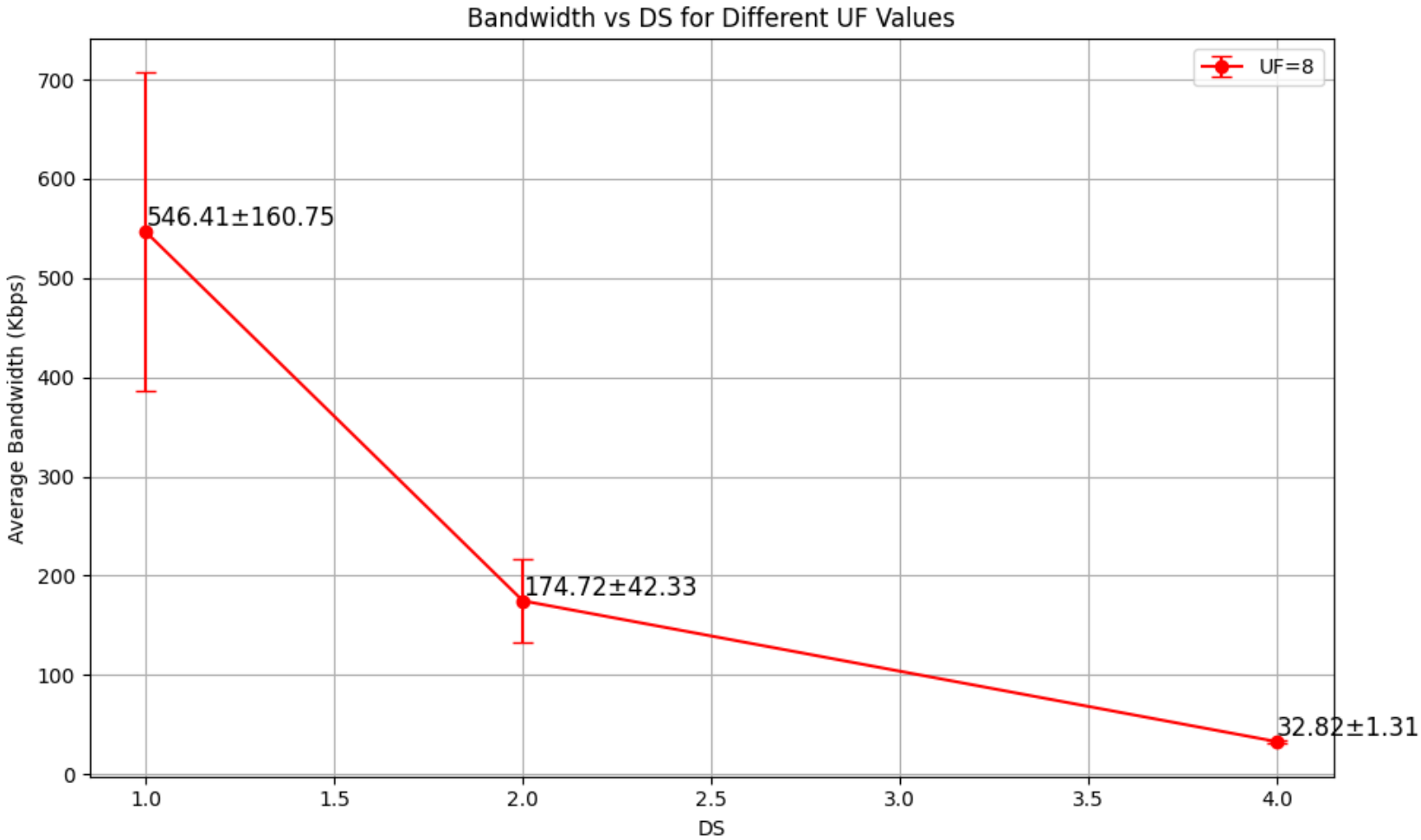

Figure: BW vs DS

Results & Observations

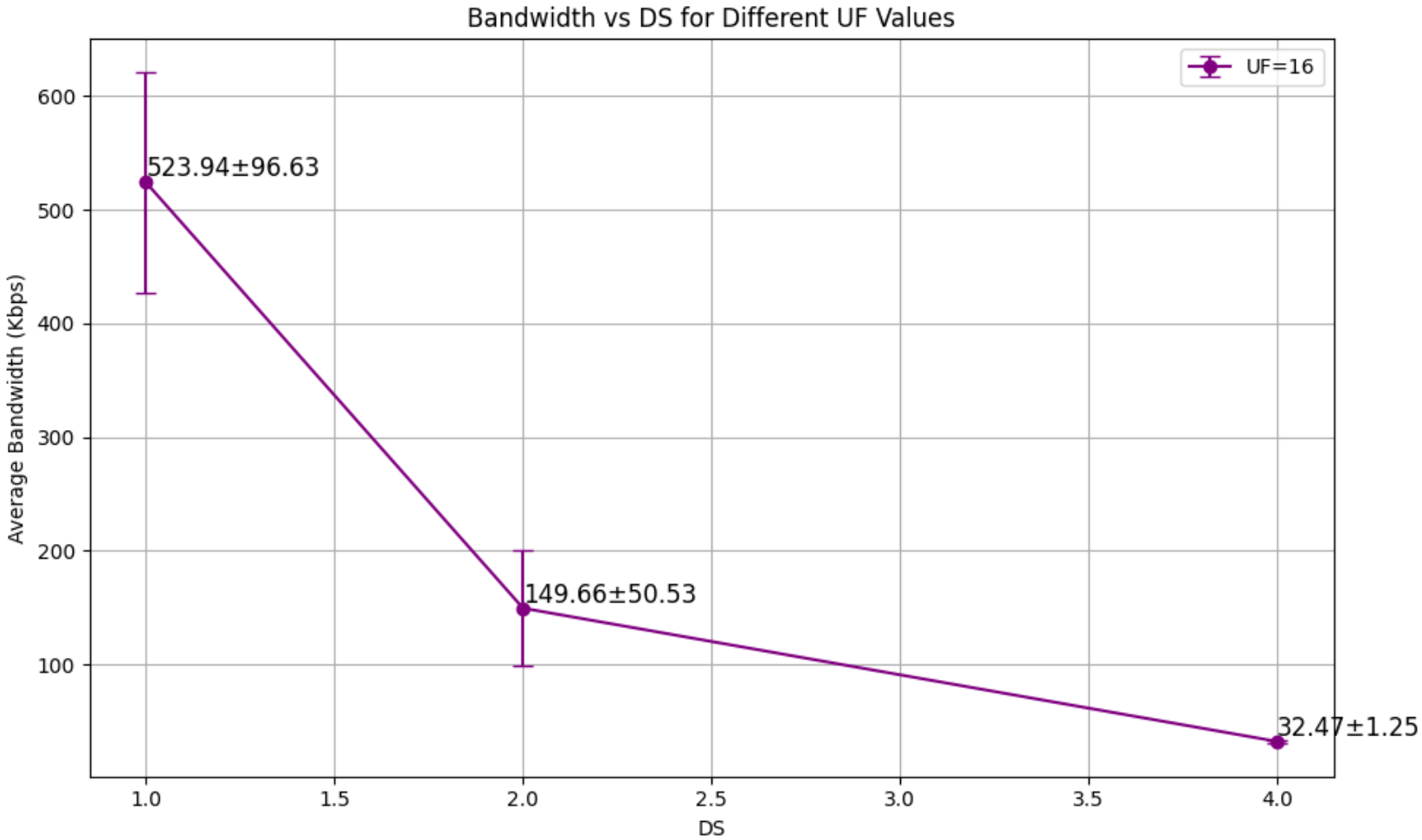

Figure: BW vs DS

Results & Observations

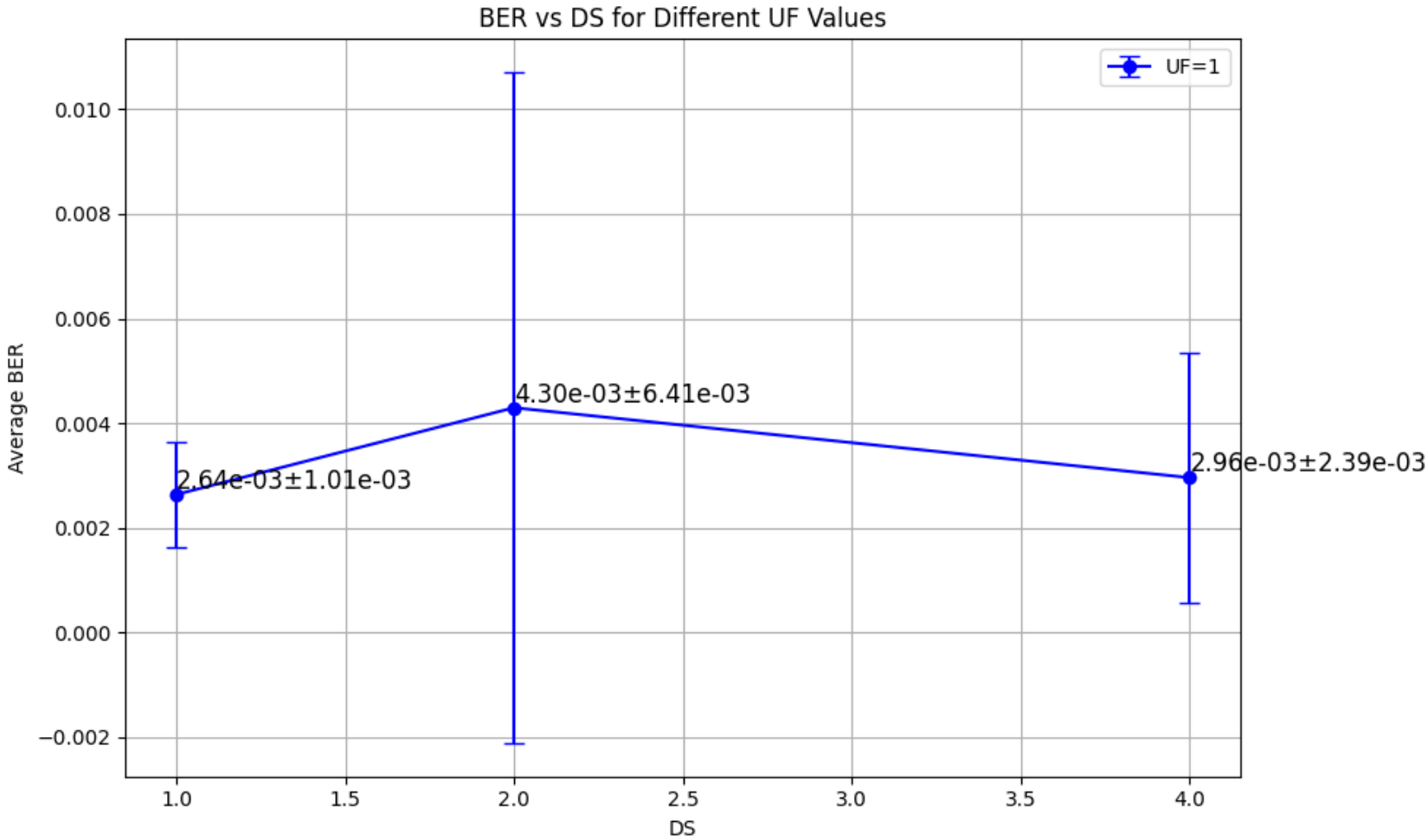

Figure: BER vs DS

Results & Observations

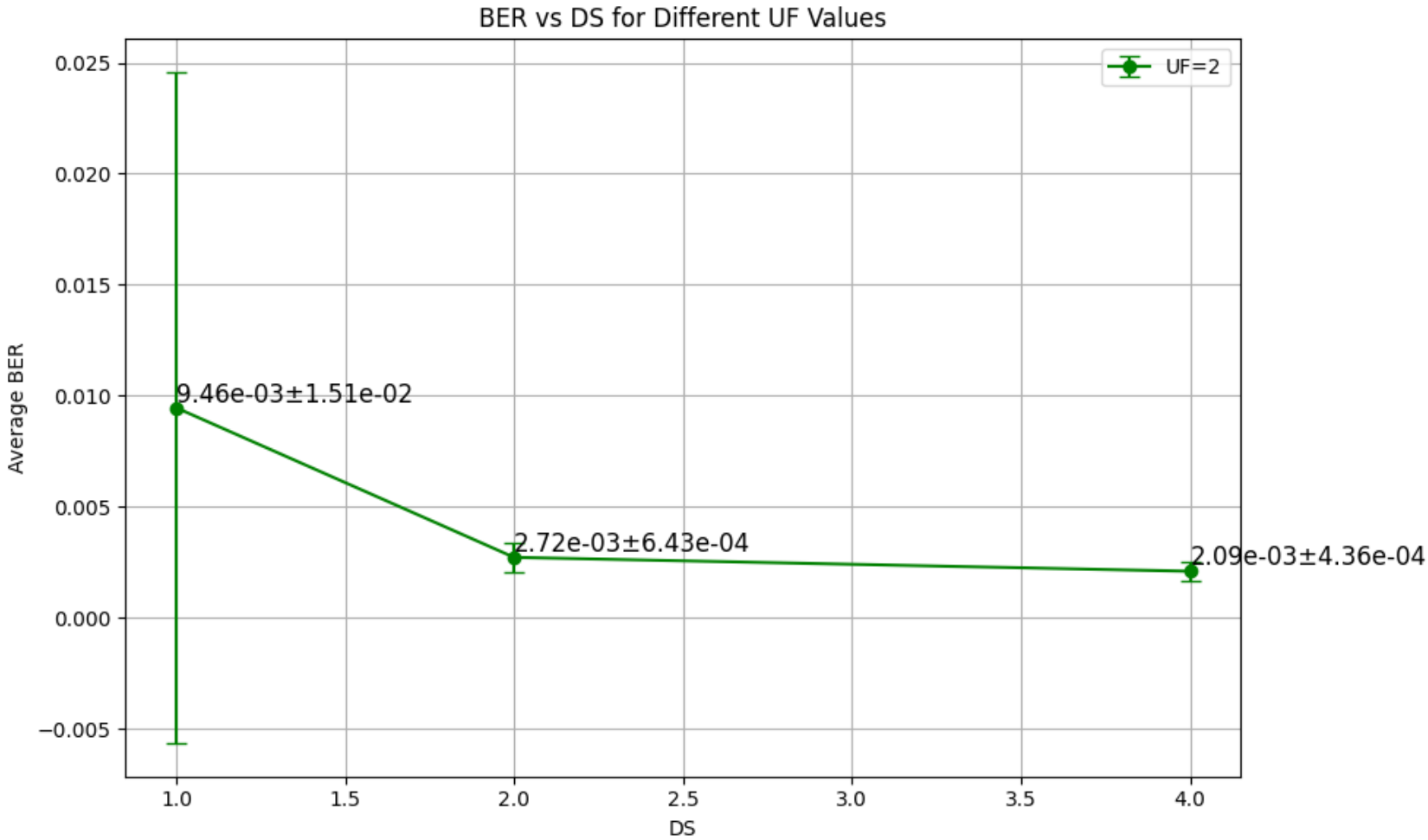

Figure: BER vs DS

Results & Observations

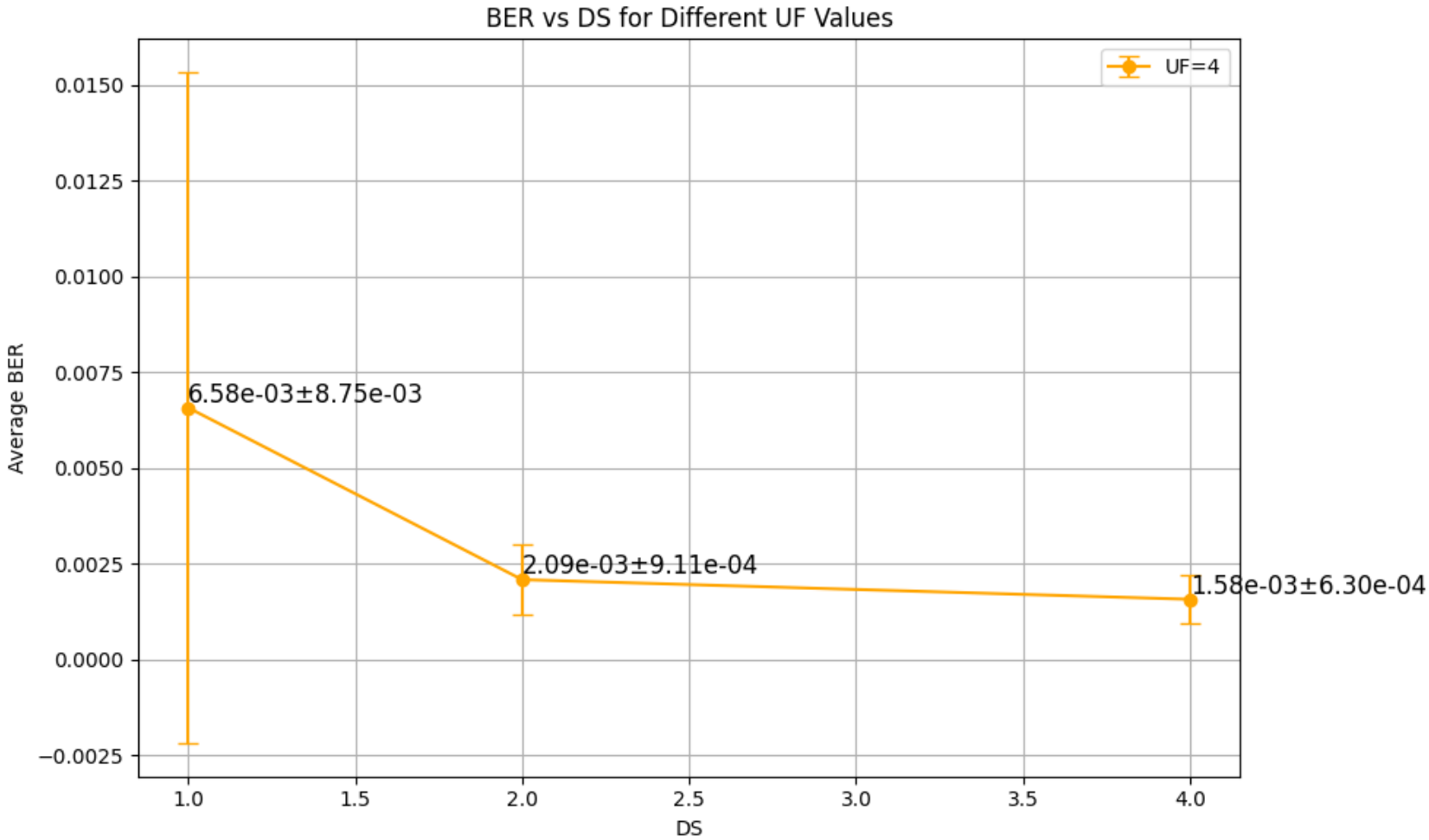

Figure: BER vs DS

Results & Observations

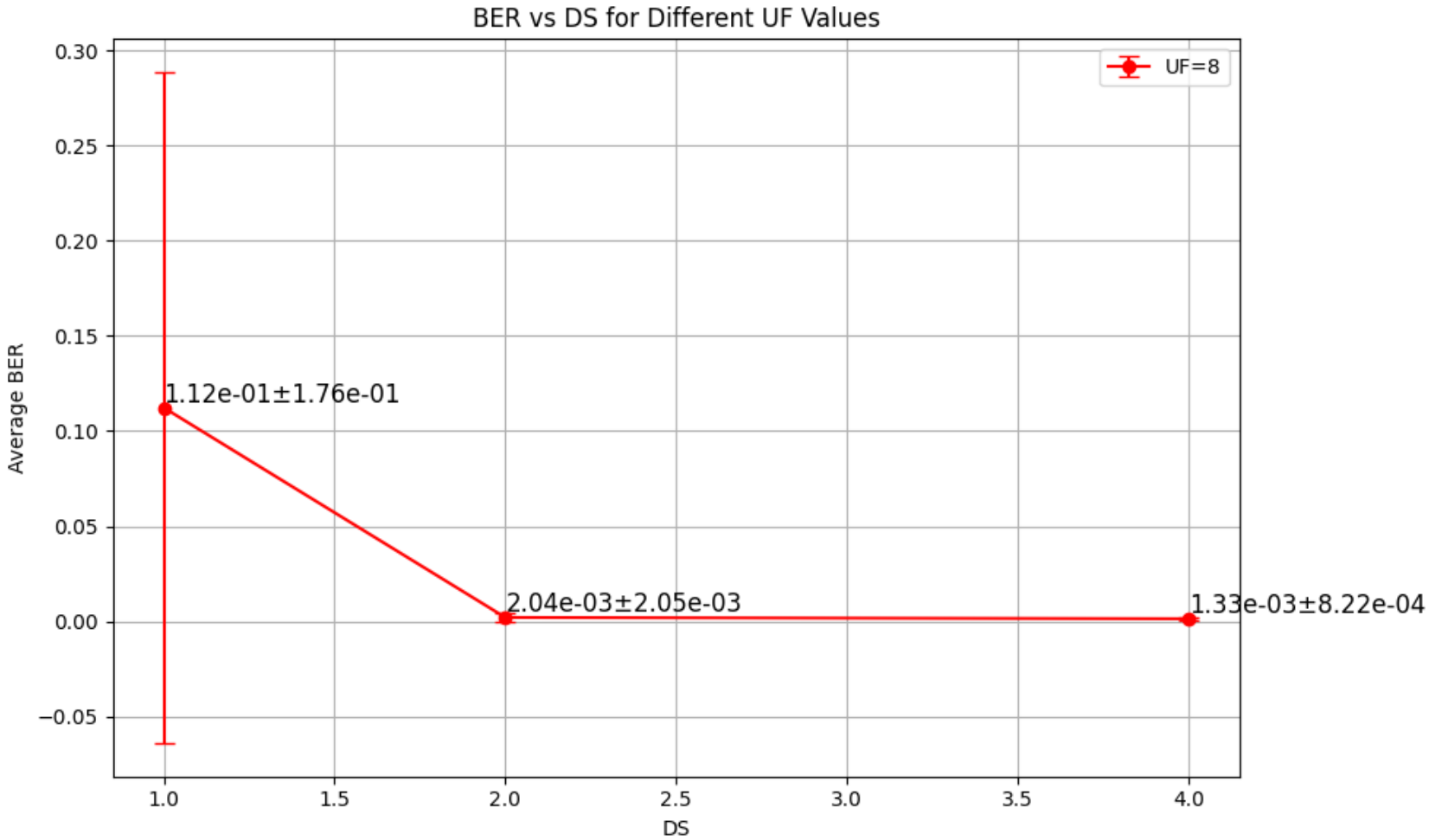

Figure: BER vs DS

Results & Observations

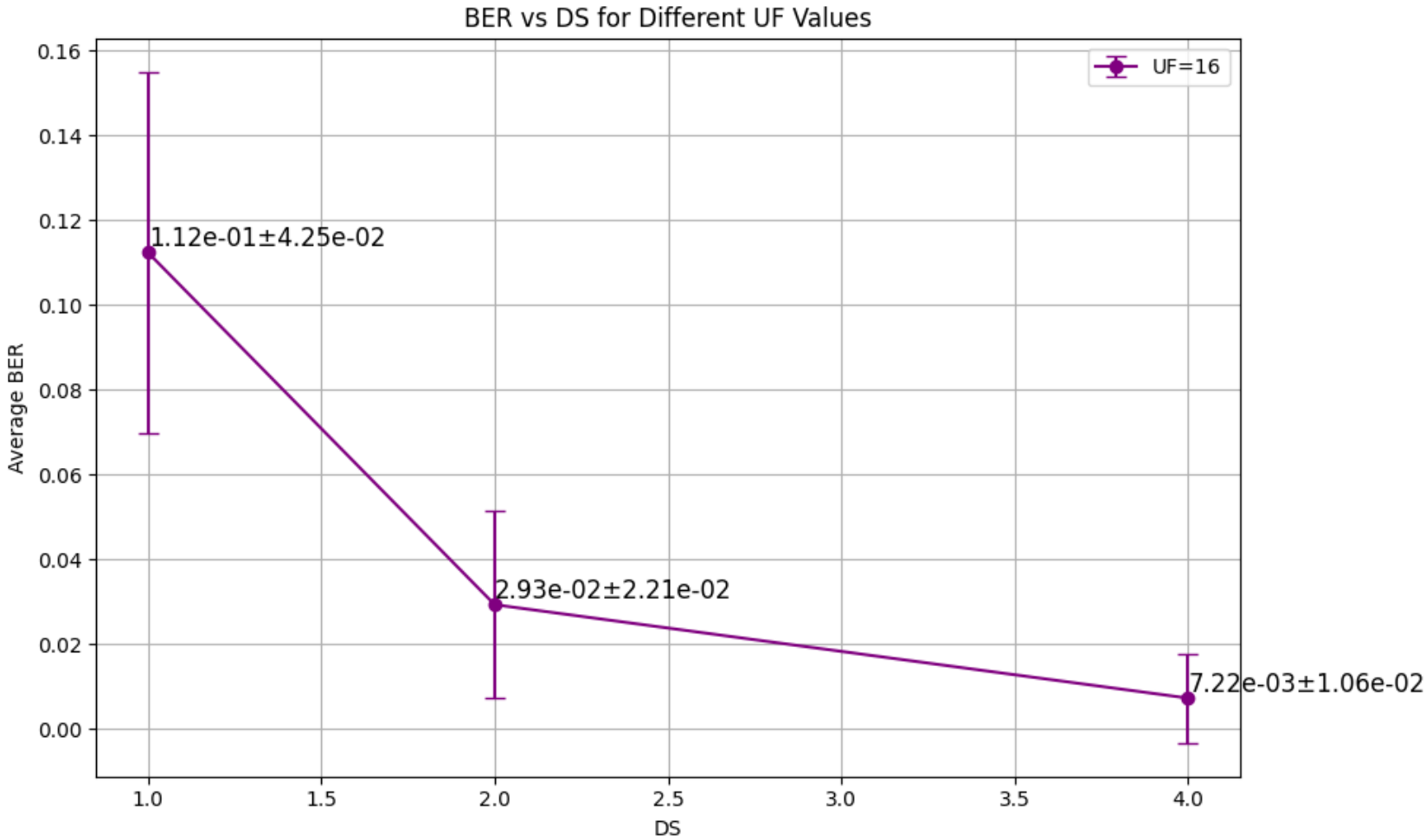

Figure: BER vs DS

Results & Observations

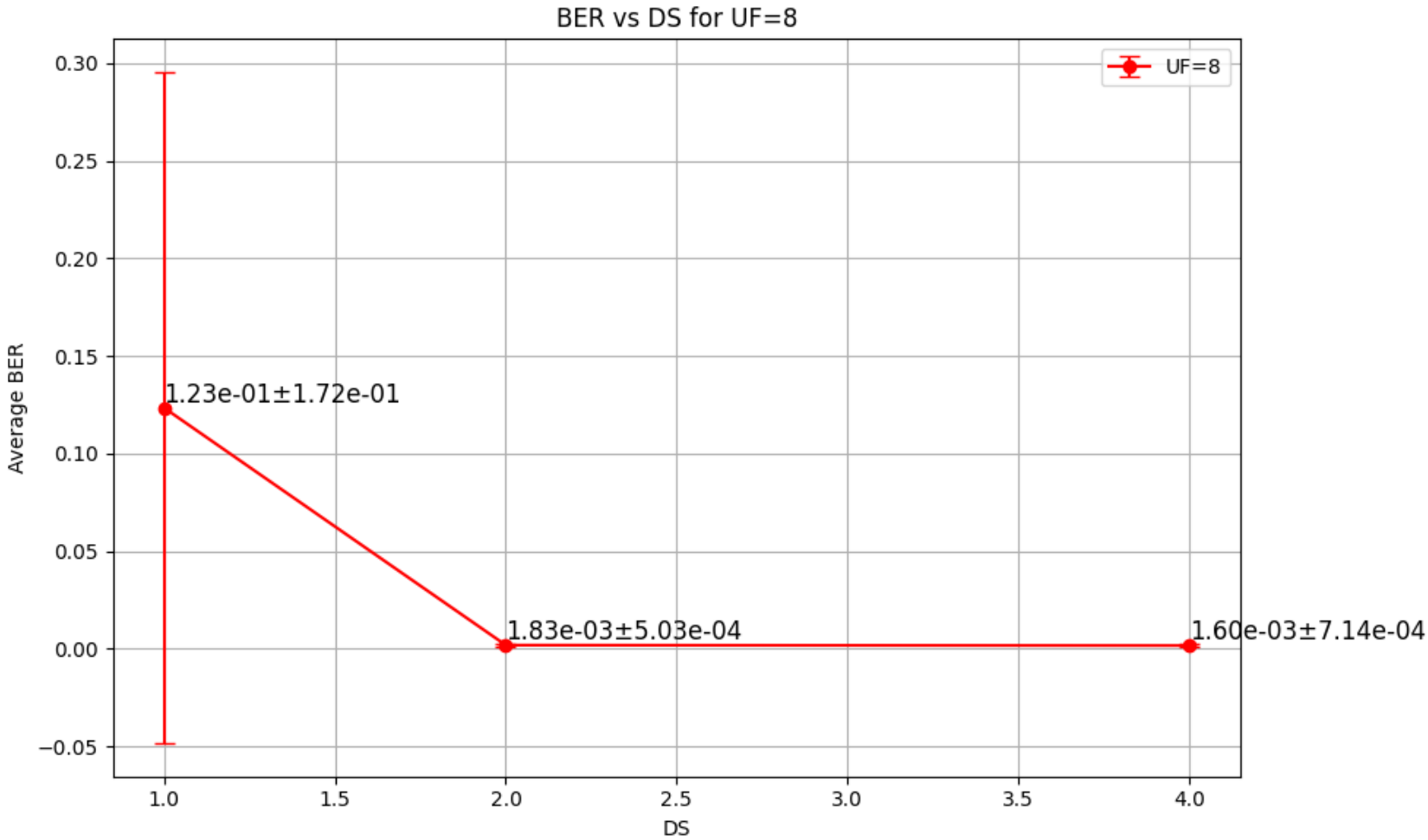

Figure: BER vs DS

Results & Observations

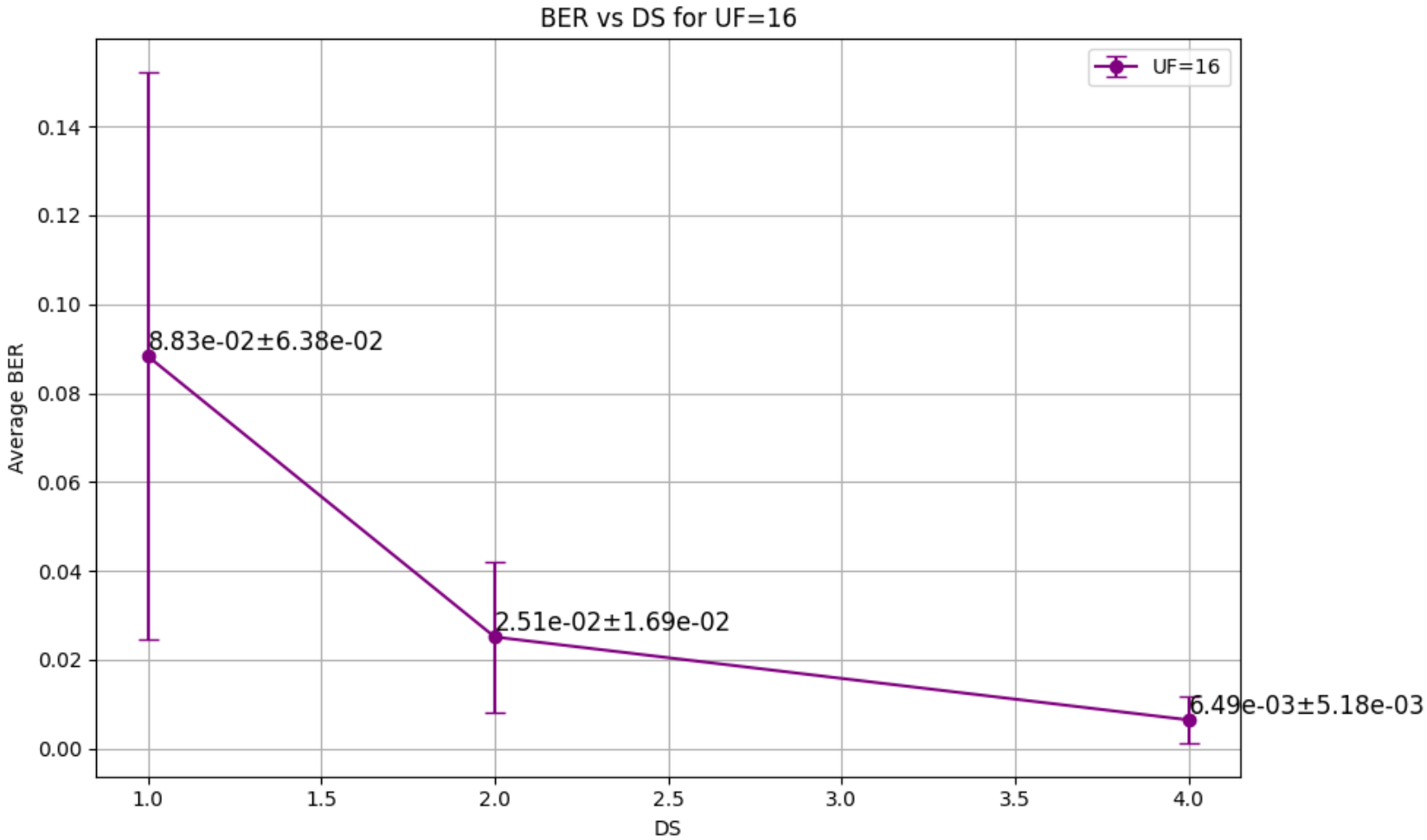

Figure: BER vs DS